



Bahir Dar University
Bahir Dar Institute of Technology
Faculty of Computing
Requirement Analysis Document (RAD)

for
Industrial project on Multi-modal AI Integrity Suite: Deepfake Detection, Content Authentication, and Intelligent CV Generation System

Submitted to the faculty of computing in partial fulfillment of the requirements for the degree
of Bachelor of Science in Software Engineering

Group Members:

Bezawit Megersa

BDU1405067

Rebika Yhinew

BDU1405547

Melat Mekonnen

BDU1405385

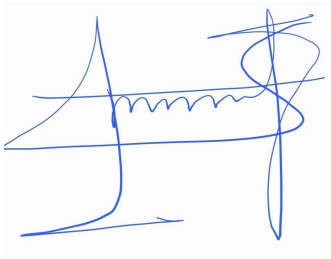
Advisor: Mr. Tesfahun N.

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Bahir Dar University, Bahir Dar Institute of Technology, Bahir Dar, Ethiopia

APPROVAL SHEET

Approved by Advisor



Mr. Tesafahun Nurrie

Full Name

Signature

March/2/2026

Date

Approved by Board of Examiners

Full Name

Signature

Date

Full Name

Signature

Date

Full Name

Signature

Date

Declaration

The Project is our own and has not been presented for a degree in any other university and all the sources of material used for the project have been duly acknowledged.

Name

Signature

Name

Signature

Name

Signature

Program: Software Engineering

Project Title: Multi-modal AI Integrity Suite: Deepfake Detection, Content Authentication, and Intelligent CV Generation System

This is to certify that I have read this project and that in my supervision and the students’ performance, it is fully adequate, in scope and quality, as a project for the degree of Bachelor of Science.

Name of Advisor

Signature

Examiner Committee Members	Signature	Date
• Examiner 1	_____	_____
• Examiner 2	_____	

It is approved that this project has been written in compliance with the formatting rules laid down by the faculty.

Roles and Responsibilities of the Group Members

List of Tasks	List members		
	Bezawit Megersa	Rebika Yhinew	Melat Mekonnen
Documentation	✓	✓	✓
Data Collecting	✓	✓	✓
Data Training		✓	
Data Cleaning	✓	✓	
UI Design	✓		
Frontend Implementation	✓	✓	
Backend Implementation		✓	✓

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List of acronyms

- **RAD:** Requirements Analysis Document
- **CV:** Convolutional Neural Network
- **AI-Chatbot:** Artificial Intelligence–based Chatbot
- **JWT :** JSON Web Token
- **CNN:** Convolutional Neural Network
- **AI:** Artificial Intelligence
- **MP4:** MPEG-4 Video Format
- **OTP:** One-Time Password
- **ML:** Machine Learning
- **DL:** Deep Learning
- **API:** Application Programming Interface
- **UI:** User Interface
- **UX:** User Experience
- **PDF:** Portable Document Format
- **IDE:** Integrated Development Environment
- **CRC:** Class-Responsibility-Collaborator
- **OpenCV:** Open-Source Computer Vision Library
- **JSON:** JavaScript Object Notation for Data Exchange
- **HTML:** Hypertext Markup Language
- **GDPR:** General Data Protection Regulation
- **GPU:** Graphics Processing Unit

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Chapter One

1.1 Background

Artificial Intelligence (AI) has become one of the most transformative technologies of the modern digital era. Recent advancements in deep learning, computer vision, and natural language processing have enabled machines to generate highly realistic multimedia content, including images, videos, and text. While these innovations have significantly improved productivity and creativity, they have also introduced new challenges related to authenticity, trust, and academic integrity.

One of the most notable developments in AI is the emergence of generative models, such as Generative Adversarial Networks (GANs), diffusion models, and large language models (LLMs). These technologies make it possible to generate realistic videos, manipulate images, and produce human-like written content. As a result, distinguishing between genuine and artificially generated media has become increasingly difficult.

Deepfake technology, which uses deep learning techniques to manipulate video or image content, has gained widespread attention due to its potential misuse. Fake media can be used to spread misinformation, impersonate individuals, manipulate public opinion, or damage reputations. At the same time, AI-generated academic content has raised concerns in educational institutions, where students may submit machine-generated assignments that are difficult to detect.

In addition, universities often face challenges in maintaining academic originality, as there is often no centralized system to compare newly submitted projects with previously submitted works. This creates a risk of plagiarism and duplication of academic research. Furthermore, many students lack access to intelligent tools that can assist them in generating professional documents such as curriculum vitae (CVs) for career opportunities.

To address these challenges, this project proposes the development of a Multi-modal AI Integrity Suite, an integrated system designed to detect manipulated media, verify academic originality, and support professional document generation. The system combines techniques

from computer vision, natural language processing, and similarity analysis to analyze multiple types of digital content.

By providing a centralized web-based platform capable of detecting deepfake videos and images, identifying AI-generated text, checking project similarity, and generating professional CVs, the proposed system aims to strengthen digital trust, support academic integrity, and assist students in their professional development.

1.2 Statement of the Problem

The rapid growth of generative artificial intelligence has introduced serious challenges in maintaining digital authenticity and academic integrity. Modern AI systems are capable of generating highly realistic videos, images, and written text that are difficult to distinguish from genuine human-created content.

Deepfake technologies allow malicious users to manipulate videos and images, which can be used to spread misinformation, impersonate individuals, and damage reputations. Similarly, AI-powered text generation tools enable users to produce large amounts of written content that may appear authentic but is actually machine-generated.

Educational institutions are particularly affected by these developments. Students may submit AI-generated assignments or reuse previously submitted academic projects, making it difficult for instructors to verify originality. Currently, many universities lack automated systems that can compare student submissions against previous projects or detect AI-generated academic content.

Additionally, students often struggle to create professional CVs that effectively present their skills and experiences. While various online tools exist, many are either complex, expensive, or not tailored for student use.

The absence of a centralized platform capable of addressing these issues simultaneously creates significant risks to digital trust, academic integrity, and professional development.

Therefore, the core problem addressed in this project is the lack of an integrated AI-based system capable of:

- Detecting manipulated videos and images

- Identifying AI-generated text
- Checking similarity between academic projects
- Assisting students in generating professional CVs

This project aims to address these challenges by designing and developing a Multi-modal AI Integrity Suite that integrates multiple AI technologies into a single web-based platform.

1.3 Objectives

1.3.1 General Objective

The general objective of this project is to design and develop a Multi-modal AI Integrity Suite that detects manipulated media, verifies academic originality, and assists users in generating professional CVs.

1.3.2 Specific Objectives

The specific objectives of the project are:

- To study existing techniques for deepfake detection, AI text detection, and similarity analysis.
- To collect and preprocess datasets for training models related to video, image, and text analysis.
- To develop AI models capable of detecting manipulated videos and images.
- To design and implement natural language processing models that can identify AI-generated academic text.
- To design a similarity checking mechanism that compares student projects with previously submitted works.
- To create an automated CV generation module based on user-provided information.
- To develop a user-friendly web-based platform that integrates all system functionalities.
- To test all developed models and system components for accuracy, reliability, and performance.
- To deploy the integrated platform in a live environment, ensuring accessibility and scalability for end-users.

1.4 Methodology

1.4.1 Requirement Gathering Methods

The requirements for the system were gathered using several methods:

1.Literature Review

Research papers, journals, and academic publications related to deepfake detection, AI text detection, and academic similarity systems were studied to understand existing technologies and limitations.

An overview of multimodal AI systems that integrate visual, audio, and textual cues to improve deepfake detection. This paper [1] discusses why multimodal approaches are stronger than single-modality methods in spotting manipulated media

A survey paper[2] that examines multimodal frameworks combining video, audio, and text, highlighting cross-modal consistency and semantic correlation for detecting deepfakes.

This research proposes a multimodal fake news classifier using contrastive learning and large language models to jointly analyze text and image features a useful reference for systems blending modalities. [3]

A systematic review [4] of AI-based methods for detecting fake news and deepfake content, including multimodal strategies and explainable AI considerations.

2.Document Analysis

Existing systems, datasets, and open-source AI tools were analyzed to determine functional and non-functional system requirements.

3.Expert Consultation

Discussions were conducted with instructors, AI researchers, and developers to gain insights into effective approaches for implementing multi-modal AI systems.

An object-oriented requirement modeling approach was adopted to structure system functionality and improve modular system design.

1.4.2 Analysis and Design Methodology

The system follows the Object-Oriented Analysis and Design (OOAD) methodology. This approach models the system as interacting components and objects, allowing better modularity, maintainability, and scalability.

The system design is represented using Unified Modeling Language (UML) diagrams including:

- Use Case Diagrams
- Class Diagrams
- Sequence Diagrams
- Activity Diagrams
- State Chart Diagrams

Tools such as StarUML and Draw.io are used to design and visualize system architecture.

1.4.3 Implementation Methodology

The implementation of the Multi-modal AI Integrity Suite utilizes modern AI and web development technologies to ensure high performance, scalability, and efficient processing.

Category	Technology	Reason for Selection
Programming Language	Python	Chosen for its simplicity and extensive support for AI and machine learning development.
Deep Learning Frameworks	TensorFlow and PyTorch	Provide powerful and flexible tools for building and training deep learning models with high accuracy.

Computer Vision Libraries	OpenCV, Dlib, and MTCNN	Enable efficient image processing, face detection, and feature extraction required for media analysis.
Natural Language Processing	Transformer-based NLP models	Selected for their high performance in understanding and classifying text, especially for AI-generated content detection.
Backend Framework	Node.js	Uses a non-blocking, event-driven architecture that efficiently handles multiple concurrent requests.
Database System	MongoDB	Offers a flexible NoSQL structure suitable for storing unstructured and semi-structured data.
Frontend Technologies	Next.js, TypeScript, and Tailwind CSS	Ensure a fast, responsive, and scalable user interface with improved developer productivity.
Development Environment	Kaggle and Visual Studio Code	Provide strong support for model training, experimentation, and efficient code development.

Table 1:Implementation Methodology

1.5 Feasibility Study

1.5.1 Economic Feasibility

The proposed system is economically feasible as it does not require significant financial investment for access or usage. Being a web-based application, it can be used through commonly available devices such as smartphones, laptops, or desktop computers with

internet connectivity. This eliminates the need for specialized hardware or expensive software.

In addition, the use of open-source technologies reduces development and operational costs, making the system affordable for institutions such as universities and organizations. This ensures that the solution can be adopted and maintained without imposing a heavy financial burden.

1.5.2 Technical Feasibility

The system is technically feasible as it is designed to operate on widely available technologies and standard web browsers. The interface is simple and intuitive, allowing easy interaction without requiring advanced technical skills.

All complex processing tasks, including AI-based analysis and data computation, are handled on the server side. This reduces the processing requirements on client devices and ensures smooth performance even on devices with limited computational power. The system's compatibility with different devices and platforms further enhances its practicality.

1.5.3 Time Feasibility

The system is feasible in terms of time efficiency, as it is designed to provide fast and automated results. Tasks such as analyzing multimedia content, checking similarity, and generating CVs are completed within a short time, significantly reducing manual effort.

By automating processes that would otherwise require considerable time, the system improves efficiency and allows tasks to be completed more quickly. This makes it suitable for environments where timely results are important, such as academic and professional settings.

1.6 Significance of the Project

The Multi-modal AI Integrity Suite provides several important benefits to different stakeholders.

- **Educational Institutions:** The system helps universities maintain academic integrity by detecting AI-generated content and checking similarity between student projects.

- **Students:** Students benefit from the CV generation tool that assists them in creating professional resumes for career opportunities.
- **Researchers:** The system provides a practical platform for studying AI applications in multimedia detection and academic verification.
- **Digital Media Platforms:** The deepfake detection component helps identify manipulated media, reducing misinformation and improving digital trust.
- **Any users:** By detecting fake media and promoting responsible AI usage, the system contributes to a safer digital environment.

Overall, the project demonstrates how AI technologies can be integrated to solve real-world problems related to media authenticity, academic integrity, and professional development.

1.7 Limitations of the Project

Despite its capabilities, the system has several limitations.

The accuracy of AI models depends on the quality and diversity of training datasets. Newly developed manipulation techniques may not always be detected with complete accuracy.

Training deep learning models requires significant computational resources, which may affect performance on systems with limited hardware capabilities.

Due to time constraints, the system focuses mainly on detection and analysis tasks, and some advanced features such as large-scale deployment, real-time monitoring, or advanced multimodal fusion may not be fully implemented.

1.8 Scope of the Project

The project focuses on the development of a web-based platform that integrates multiple AI functionalities. The system includes features such as deepfake detection for videos, deepfake detection for images, detection of AI-generated academic text, academic project similarity checking, automated CV generation, and a web-based dashboard for system interaction.

The system is designed primarily for academic and experimental use and demonstrates the integration of AI technologies for solving real-world problems. Features such as large-scale

deployment, real-time media monitoring, and integration with external educational platforms are considered outside the current project scope.

1.9 Organization of the Project

This document is organized into several chapters that describe the development of the proposed system.

Chapter One: introduces the project, including the background, problem statement, objectives, methodology, feasibility analysis, significance, limitations, and scope.

Chapter Two: presents the system features and requirement analysis, including the existing system, proposed system, functional requirements, non-functional requirements, and UML diagrams.

Chapter Two: System Features

2. The Existing System

The verification of digital content authenticity has become an increasingly complex challenge due to the rapid development of generative artificial intelligence technologies. Existing systems designed to detect manipulated media or verify digital content authenticity are often fragmented and limited in scope. Most current solutions focus only on a single type of content, such as video deepfake detection or plagiarism detection, rather than providing an integrated solution capable of analyzing multiple content formats.

At present, the detection of manipulated media and AI-generated content typically relies on a combination of manual verification processes, limited automated detection tools, and proprietary systems deployed by large technology platforms. While these approaches provide partial protection against manipulated media and misinformation, they suffer from several limitations including inefficiency, lack of transparency, limited scalability, and restricted accessibility.

Additionally, many educational institutions lack automated systems capable of verifying the originality of student academic work or detecting AI-generated text. Students also have limited access to intelligent tools that assist them in creating professional documents such as resumes or curriculum vitae (CVs). As a result, existing systems fail to address the broader challenges of digital media authenticity, academic integrity, and intelligent document generation in a unified platform.

2.1 Existing System Components

2.1.1 Manual Verification Systems

One of the most common approaches used for detecting manipulated media is manual verification performed by human reviewers. Journalists, social media moderators, cybersecurity analysts, and law enforcement personnel often inspect videos, images, or documents to determine whether they have been manipulated or artificially generated.

During manual analysis, reviewers examine several visual and contextual cues. For example, in video verification, analysts may observe facial expressions, lip synchronization, eye movements, lighting inconsistencies, and unusual visual artifacts that may indicate deepfake manipulation. Similarly, when verifying images, reviewers may examine irregular textures, inconsistent shadows, or distorted facial structures that could signal synthetic image generation.

In critical situations such as legal investigations or high-profile media verification, digital forensic experts may analyze multimedia files using specialized forensic software tools. These tools can examine metadata, compression patterns, or frame-level artifacts within a video. While such forensic techniques can provide highly accurate results when conducted by experienced professionals, they require significant technical expertise and are often time-consuming.

Because manual verification relies heavily on human judgment, it is difficult to apply this method to large volumes of digital content. With the rapid growth of social media platforms and online content sharing, thousands of videos and images are uploaded every minute. As a result, manual verification alone is no longer sufficient to detect manipulated media efficiently or in real time.

2.1.2 Semi-Automated Detection Tools

To reduce the burden of manual verification, several semi-automated detection tools have been developed. These tools typically allow users to upload digital media such as videos or images to an online platform where automated algorithms perform authenticity checks.

Many of these platforms analyze visual patterns, compression artifacts, or metadata associated with multimedia files. For example, some systems examine frame rates, encoding signatures, or file history data to identify potential manipulation. Other tools use basic machine learning models trained on limited datasets to classify content as real or synthetic. Although semi-automated detection tools provide faster analysis compared to manual verification, they often operate as black-box systems, meaning they provide limited explanations regarding how the final decision was made. Users may receive a simple result indicating whether the content is likely real or fake, without clear reasoning or confidence scores.

Another limitation of these tools is their inconsistent accuracy. Because many models are trained using small or specialized datasets, they may perform well in controlled environments but fail when applied to real-world content that varies in lighting conditions, resolution, compression levels, or video quality. Additionally, metadata-based detection techniques are unreliable because metadata can easily be modified or removed during file processing.

2.1.3 Platform-Level Automated Systems

Large technology companies and social media platforms have begun deploying automated systems designed to detect manipulated media and misinformation. These systems use advanced artificial intelligence models to analyze content uploaded to their platforms and flag potentially manipulated media. While these systems represent a significant technological advancement, they have several important limitations. First, the algorithms used by these platforms are typically proprietary and not publicly accessible. This lack of transparency prevents researchers, journalists, and independent organizations from evaluating the accuracy or reliability of the detection methods. Second, these systems are often designed to work specifically within the platform where they are deployed. As a result, their functionality cannot easily be integrated into independent applications or academic research projects. Finally, even advanced automated detection systems may struggle when encountering new deepfake generation techniques. Modern generative models such as GANs and diffusion-based architectures continue to evolve rapidly, producing increasingly realistic synthetic media. Detection models trained on older datasets may therefore fail to recognize newly generated manipulation techniques.

Step-by-Step Operation of the Existing System

In general, the workflow of existing deepfake detection or media verification systems follows several basic stages. Initially, digital content is either uploaded to a platform or discovered through monitoring systems. In some cases, users may report suspicious media, prompting further investigation. Once the content is identified, preliminary screening may be conducted either manually or through automated heuristics that attempt to detect obvious signs of manipulation. After the initial screening process, the content undergoes a deeper analysis phase. This may involve manual inspection by human reviewers, analysis using semi-automated detection tools, or evaluation using AI-based algorithms deployed by large

platforms. Based on the outcome of this analysis, a decision is made regarding the authenticity of the content.

Finally, an action is taken depending on the evaluation results. The content may be labeled as manipulated, removed from the platform, or allowed to remain if no clear evidence of manipulation is detected. In many cases, the lack of clear explanations or standardized confidence scores can make these decisions difficult and inconsistent.

Drawbacks of the Existing System

Although current systems provide some level of protection against manipulated media, they suffer from several significant limitations. One major limitation is the lack of scalability. Manual verification and semi-automated detection approaches cannot keep pace with the enormous volume of digital content generated daily across social media platforms and online environments. As a result, many manipulated videos and images remain undetected.

Another challenge is limited detection accuracy. Many existing systems struggle to detect highly realistic deepfakes generated using advanced artificial intelligence models. As generative AI technologies continue to improve, distinguishing between real and synthetic media becomes increasingly difficult. Existing detection systems also suffer from poor generalization. Models trained using limited datasets may perform well in controlled testing environments but fail when exposed to real-world content that contains variations in lighting, compression, camera quality, or motion patterns. Transparency and explainability also remain significant concerns. Many systems provide limited information about how their decisions are made, which reduces user trust and makes it difficult to interpret results.

Finally, accessibility remains restricted in many cases. Several advanced detection tools are proprietary systems owned by large organizations, meaning they are not easily available to independent researchers, academic institutions, or the general public. Because of these limitations, there is a strong need for an integrated solution capable of analyzing multiple types of digital content, improving detection accuracy, and providing transparent and accessible verification tools.

2.2 Proposed System

The proposed system, titled Multi-modal AI Integrity Suite: Deepfake Detection, Content Authentication, and Intelligent Document Generation System, is designed as an intelligent and integrated platform that addresses the limitations of existing digital content verification solutions. Unlike traditional systems that focus only on a single type of media analysis, the proposed solution combines multiple artificial intelligence technologies into a unified platform capable of analyzing different forms of digital content.

The system is designed to detect manipulated media, verify the authenticity of academic content, and assist users in generating professional documents. By integrating computer vision, natural language processing, and similarity detection algorithms, the system provides a comprehensive solution for ensuring both digital integrity and academic honesty.

The platform enables users to upload and analyze different types of content, including videos, images, and text documents. Video and image content are analyzed using deep learning–based computer vision models that identify artifacts commonly introduced during synthetic media generation. For textual content, natural language processing techniques are applied to detect patterns that indicate AI-generated text or similarities with previously submitted academic work. Additionally, the system includes an intelligent CV generation module that allows users to generate professional resumes based on structured input.

The proposed system is implemented as a web-based platform to ensure accessibility and ease of use. Users can interact with the system through an intuitive interface that supports content upload, analysis requests, result visualization, and document generation. The architecture is designed to be modular and scalable, allowing different AI modules to operate independently while communicating through a centralized backend.

Another key component of the proposed system is the integration of an AI-powered chatbot. The chatbot assists users in understanding system functionality, interpreting analysis results, and learning about deepfake technologies and content authenticity verification. By providing interactive guidance, the chatbot improves user experience, enhances system transparency, and promotes responsible use of artificial intelligence.

Overall, the proposed system aims to provide a centralized, automated, and scalable solution that strengthens digital trust, protects academic integrity, and supports students in professional development.

Key Functional Capabilities of the Proposed System

The system includes several integrated functional modules that work together to provide a comprehensive digital integrity platform. The first module focuses on media authenticity detection. Users can upload videos or images through the web interface, and the system automatically performs preprocessing and analysis using computer vision techniques. Video frames are extracted and processed using libraries such as OpenCV and FFmpeg. Facial regions are detected using advanced face detection models such as MTCNN or Dlib, allowing the system to focus on areas where deepfake artifacts commonly appear. The extracted features are then analyzed using deep neural network architectures such as XceptionNet or EfficientNet to identify inconsistencies in facial movement, texture, and lighting patterns. Based on this analysis, the system classifies the media as authentic or manipulated and provides a confidence score indicating the reliability of the prediction.

The second module supports AI-generated text detection. Academic documents uploaded by users are processed using natural language processing techniques and transformer-based models that analyze linguistic patterns, sentence structure, and statistical features of the text. This allows the system to determine whether the content was likely generated by an artificial intelligence model or written by a human author.

The third module focuses on academic similarity detection. When students upload research papers or project documents, the system compares the submitted content with previously stored academic work. Similarity detection algorithms analyze textual patterns and structural similarities to identify potential plagiarism or duplication. The results are presented as similarity scores, helping instructors and institutions maintain academic integrity.

Another important feature of the system is intelligent CV generation. Users can enter personal information, education history, technical skills, and work experience through structured forms. The system then automatically generates a professionally formatted CV based on standardized templates. This feature helps students and graduates create well-structured resumes suitable for job applications. To improve usability and transparency, the system provides an interactive result visualization dashboard. After content analysis is completed, results are displayed in a clear and understandable format, including classification labels, similarity scores, and AI detection indicators. Visual feedback helps users quickly interpret the outcome of the analysis. The integrated AI chatbot further enhances the user

experience by providing assistance during system interaction. Users can ask questions related to deepfake detection, content authenticity, or system usage. The chatbot provides explanations in natural language, helping users understand the results and the reasoning behind them.

Finally, the system architecture ensures secure and reliable operation through user authentication, secure file storage, and controlled access to analysis data. Uploaded content and generated results are stored in a database for history tracking and future reference.

Advantages of the Proposed System

The proposed system offers several advantages compared to existing solutions.

First, the system provides multi-modal analysis, meaning it can process and verify multiple types of digital content including videos, images, and text documents. This integrated approach addresses the limitations of traditional systems that focus only on one media format. Second, the use of advanced deep learning and natural language processing models improves the accuracy and reliability of detection results. The system is capable of identifying sophisticated deepfake media generated using modern AI techniques such as GANs and diffusion models. Another major advantage is the automation of the verification process. Once users upload content, the system automatically performs preprocessing, feature extraction, analysis, and result generation without requiring manual intervention. This significantly reduces the time required to verify digital content. The system also emphasizes transparency and interpretability. Confidence scores, similarity metrics, and chatbot explanations help users understand how the system reached its conclusions, which increases trust in the results. From a usability perspective, the platform is designed with a user-friendly web interface that allows even non-technical users to upload content, view results, and generate professional documents easily.

Additionally, the modular design of the system allows for future expansion and scalability. New modules such as audio deepfake detection, real-time media analysis, or additional document verification tools can be integrated without major changes to the system architecture.

2.3 Requirement Analysis

Requirement analysis defines the functional capabilities that the proposed system must provide in order to meet user needs and achieve the objectives of the project.

2.3.1 Functional Requirements

The system must allow users to create accounts using basic information such as name, email, and password, and enable secure login to access the dashboard. Once authenticated, users can upload various types of content, including videos and images for deepfake detection and academic documents for AI text detection and plagiarism analysis, with all files validated for format and size before processing. The system analyzes video content by extracting frames, detecting facial regions, and applying deep learning models to classify it as real or manipulated with a confidence score, while image analysis uses convolutional neural networks to detect artificial generation or manipulation, and text analysis examines linguistic patterns to identify AI-generated content and compare documents for similarity or plagiarism. Additionally, the system provides a CV generation feature that allows users to input personal and professional details and generate, edit, and download structured CVs. All results, including detection outcomes and similarity scores, are displayed clearly on a dashboard, and the system stores analysis history for future access. It also includes an AI-powered chatbot to assist users, explain results, and guide system usage, along with error handling for invalid inputs and processing failures. Furthermore, the platform may offer a community engagement feature where users participate in identifying real versus AI-generated content to promote awareness and interaction.

Main Functional Requirements

- The system allow users to create accounts using name, email, and password.
- The system allow registered users to log in securely and access the dashboard.
- The system provide logout functionality and password reset via email verification or OTP.
- The system allow users to upload videos and images for deepfake detection.
- The system allow users to upload academic documents for AI text detection and plagiarism/similarity analysis.
- The system validate uploaded files based on format and size before processing.

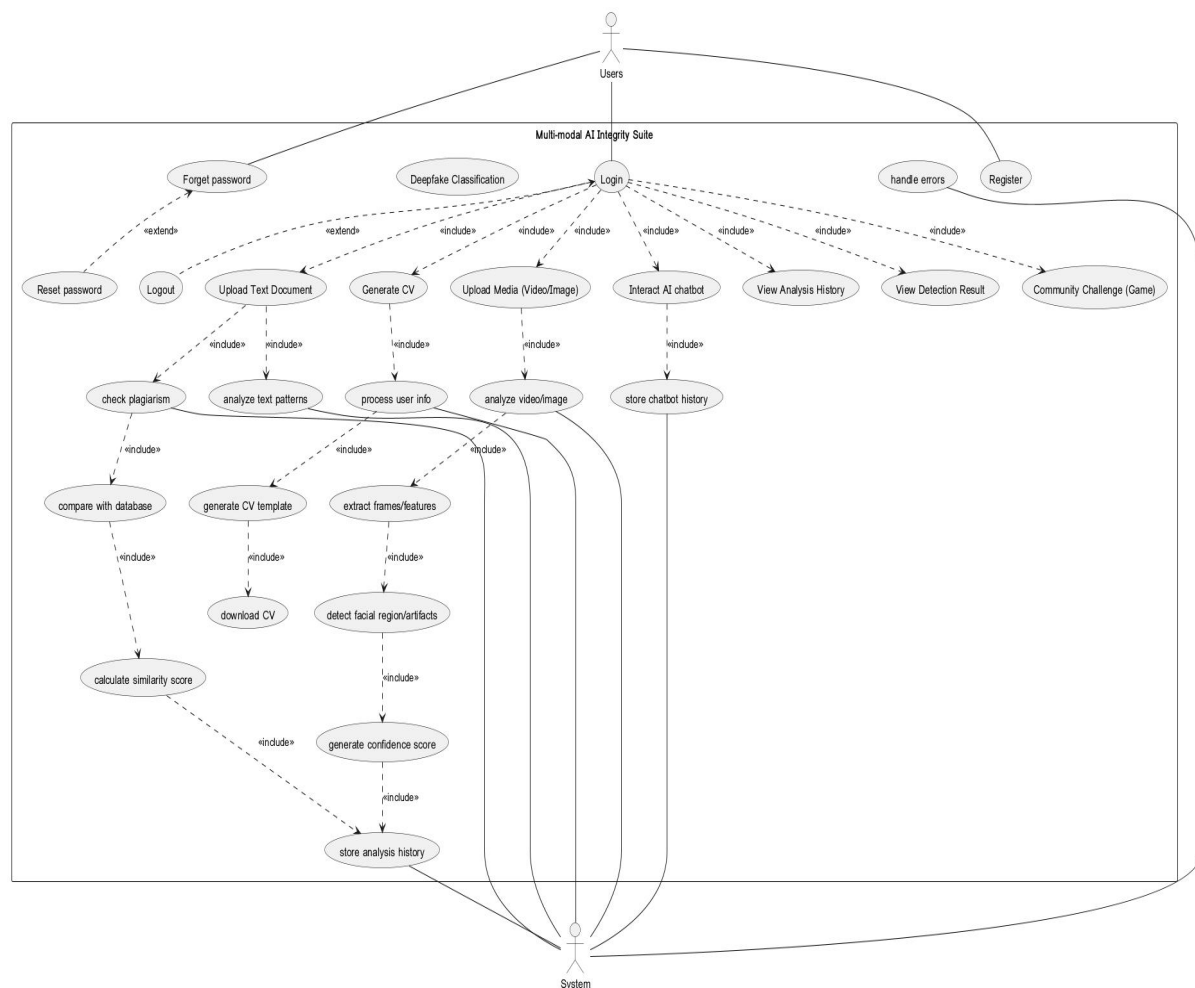
- The system process video content by extracting frames, detecting faces, and analyzing features using deep learning models.
- The system classify videos as real or manipulated and provide a confidence score.
- The system analyze images using neural networks to detect manipulation or AI generation.
- The system analyze text to detect AI-generated content using linguistic and structural patterns.
- The system compare uploaded documents with stored data to detect similarities or plagiarism.
- The system provide a CV generation feature based on user input (personal and professional data).
- The system allow users to view, edit, and download generated CVs.
- The system display analysis results clearly through a dashboard interface.
- The system present outputs such as detection results, similarity scores, and AI indicators in an understandable format.
- The system store analysis results and maintain a user history of submissions.
- The system allow users to access and review previously analyzed content.
- The system provide an AI-powered chatbot for assistance, explanations, and navigation.
- The system include error handling for invalid uploads and processing failures.
- The system include a community feature for users to identify real vs AI-generated content.

2.3.2 System Use Case

Use Case Diagram

The Multi-modal AI Integrity Suite enables users to interact with a centralized platform designed to analyze and verify different types of digital content. Through a web-based interface, users can upload videos, images, and textual documents for authenticity verification, detect AI-generated academic content, compare project submissions with previously stored research work, and generate professional curriculum vitae (CVs). The system integrates multiple artificial intelligence technologies including computer vision models for media authenticity detection, natural language processing models for text analysis,

and similarity detection algorithms for academic integrity verification. Users interact with the platform to submit content, request analysis, view results, and download reports.



Use Case Documentation

The following use case documentation provides detailed descriptions of the primary interactions between users and the Multi-modal AI Integrity Suite. Each use case describes a specific functionality of the system and outlines the sequence of events that occur when a user interacts with the platform. These use cases collectively represent the core operations of the system, including user authentication, media analysis, academic content verification, intelligent document generation, and system interaction through the chatbot.

Table 1: Create Account

Field	Description
Use Case Number	01
Use Case Name	Create Account
Description	A user who wants to use the Multi-modal AI Integrity Suite must first register by creating an account using basic identification information.
Actors	User
Entry Condition	The registration page is available and the user has a valid email address.

Flow Events	of 1. User accesses the system website. 2. System displays the registration form. 3. User enters name, email, and password. 4. User submits the form. 5. System validates the information. 6. System creates the account and stores the information in the database. 7. System redirects the user to the login page.
Exit Condition	The user account is successfully created.

Table 2:Create Account

Table 2: Login

Field	Description
Use Case Number	02
Use Case Name	Login
Description	Registered users log into the system to access content analysis and document generation services.

Actors	User
Entry Condition	The user is registered and the login page is displayed.
Flow of Events	1. System prompts user to enter email and password. 2. User enters login credentials. 3. System verifies the credentials. 4. If valid, user is logged in and redirected to the dashboard.
Alternative Flow	If credentials are incorrect, the system displays an error message and prompts the user to try again.
Exit Condition	User successfully logs into the system.

Table 3: Login

Table 3: Logout

Field	Description
Use Case Number	03
Use Case Name	Logout

Description	Allows users to securely terminate their session.
Actors	User
Entry Condition	User is currently logged into the system.
Flow of Events	1. User selects the logout option. 2. System ends the session. 3. System redirects the user to the login page.
Exit Condition	User is successfully logged out.

Table 4: Logout

Table 4: Reset Password

Field	Description
Use Case Number	04
Use Case Name	Reset Password
Description	Allows users to reset their password if they forget their login credentials.

Actors	User, System
Entry Condition	User selects the "Forgot Password" option.
Flow of Events	1. System requests registered email. 2. User enters email. 3. System sends reset link or OTP. 4. User enters new password. 5. System validates and updates password.
Exit Condition	Password is successfully updated.

Table 5:Reset Password

Table 5: Upload Media for Detection

Field	Description
Use Case Number	05
Use Case Name	Upload Media
Description	Users upload videos or images for deepfake detection and authenticity verification.

Actors	User
Entry Condition	User is logged into the system dashboard.
Flow of Events	1. User selects upload option. 2. System prompts for media file. 3. User uploads video or image. 4. System validates file format and size. 5. System stores file for processing.
Exit Condition	Media file is successfully uploaded.

Table 6: Upload Media for Detection

Table 6: Media Preprocessing

Field	Description
Use Case Number	06
Use Case Name	Media Preprocessing
Description	The system prepares uploaded media for AI analysis by extracting relevant data such as frames or features.

Actors	System
Entry Condition	Media file is successfully uploaded.
Flow of Events	1. System extracts frames from video. 2. System detects facial regions or relevant visual features. 3. Data is normalized and prepared for analysis.
Exit Condition	Media data is ready for AI analysis.

Table 7: Media Preprocessing

Table 7: Deepfake Detection

Field	Description
Use Case Number	07
Use Case Name	Deepfake Detection
Description	The system analyzes videos or images to determine whether they are real or AI-generated.

Actors	System
Entry Condition	Preprocessed media data is available.
Flow of Events	1. System applies trained AI models. 2. Model analyzes visual patterns and facial inconsistencies. 3. System classifies content as Real or Fake. 4. System generates confidence score.
Exit Condition	Detection result is generated and stored.

Table 8: : Deepfake Detection

Table 8: AI Text Detection

Field	Description
Use Case Number	08
Use Case Name	AI Text Detection
Description	Detects whether uploaded academic text is AI-generated or human-written.

Actors	User, System
Entry Condition	User uploads a text document.
Flow of Events	1. System receives text file. 2. NLP model analyzes linguistic patterns. 3. System evaluates probability of AI-generated content. 4. Result is displayed to the user.
Exit Condition	AI detection result is shown.

Table 9: AI Text Detection

Table 9: Academic Similarity Checking

Field	Description
Use Case Number	09
Use Case Name	Similarity Checking
Description	Compares uploaded academic documents with previous submissions to detect similarity.

Actors	User, System
Entry Condition	User uploads academic document.
Flow of Events	1. System stores uploaded document. 2. System compares document with stored database. 3. Similarity algorithm calculates similarity score. 4. Result is displayed.
Exit Condition	Similarity report is generated.

Table 10: Academic Similarity Checking

Table 10: CV Generation

Field	Description
Use Case Number	10
Use Case Name	Generate CV
Description	Allows users to automatically generate a professional CV using their information.

Actors	User
Entry Condition	User selects CV generation option.
Flow of Events	1. User enters personal details, education, and skills. 2. System processes the information. 3. System generates formatted CV template. 4. User downloads or edits the CV.
Exit Condition	CV is successfully generated.

Table 11: CV Generation

Table 11: View Analysis Results

Field	Description
Use Case Number	11
Use Case Name	View Results
Description	Displays the results of media analysis, text detection, or similarity checking.

Actors	User
Entry Condition	Content analysis has been completed.
Flow of Events	1. System retrieves results. 2. System displays classification labels, scores, and reports.
Exit Condition	Results are displayed to the user.

Table 12: View Analysis Results

Table 12: AI Chatbot Interaction

Field	Description
UseCase Number	12
Use Case Name	Chatbot Interaction
Description	Provides users with assistance regarding system usage and AI detection explanations.
Actors	User, System

Entry Condition	User opens chatbot interface.
Flow of Events	1. User submits question. 2. Chatbot processes request. 3. Chatbot generates response. 4. Interaction is stored.
Exit Condition	User receives assistance.

Table 13: AI Chatbot Interaction

2.3.2 Business Rule Documentation

The business requirements define the high-level needs and objectives that the Multi-modal AI Integrity Suite must satisfy in order to provide value to users, institutions, and stakeholders. These rules guide how the system processes digital content, verifies authenticity, supports academic integrity, and assists users in professional document creation. The primary goal of the system is to provide a centralized platform capable of analyzing multiple types of digital content, including videos, images, and text documents. Through the integration of artificial intelligence techniques such as computer vision and natural language processing, the system ensures reliable verification of media authenticity and academic originality.

The following business rules define the major operational requirements of the system.

- **Multi-modal Content Authenticity Assurance:**The system shall provide an automated mechanism to verify the authenticity of multiple forms of digital content including videos, images, and textual documents. This capability helps reduce the spread of manipulated media, misinformation, and AI-generated academic content.

- **Efficient Content Verification:** The system shall enable fast and reliable detection of manipulated media and AI-generated text. Rapid analysis supports timely decision-making for users such as students, researchers, journalists, and digital content moderators.
- **User Accessibility:** The system shall be accessible through a web-based platform that allows users to upload content, request analysis, and generate documents without requiring advanced technical knowledge. The interface must remain intuitive and easy to navigate for both technical and non-technical users.
- **Accuracy and Reliability:** The system shall employ artificial intelligence models such as deep learning networks and natural language processing algorithms to ensure accurate detection results. Each analysis result shall be accompanied by a confidence score to indicate prediction reliability.
- **Secure Data Handling:** The system shall ensure secure storage and handling of user information, uploaded media, and analysis results. Data privacy mechanisms must protect user data and prevent unauthorized access.
- **Transparent Result Presentation:** The system shall present analysis outcomes in a clear and understandable format. Detection results, similarity scores, and AI-generated content indicators must be displayed in a manner that allows users to easily interpret the system's conclusions.
- **Analysis History Management:** The system shall maintain records of previously analyzed content, including media files and academic documents. Users must be able to review their past analyses and track previously generated reports.
- **Academic Integrity Support:** The system shall compare uploaded academic documents with previously stored project submissions to identify potential similarities or duplication. This capability helps institutions maintain academic honesty and detect possible plagiarism.
- **Professional Document Generation Support:** The system shall assist users in generating professional curriculum vitae (CVs) based on structured input. The generated CVs must follow standardized formatting suitable for academic and professional job applications.
- **Intelligent User Assistance:** The system shall include an AI-powered chatbot that provides guidance to users regarding system usage, interpretation of detection results, and general questions about deepfake technologies or AI-generated content.

- **Scalability and System Expansion:** The architecture of the system shall support future expansion and integration of additional modules such as audio deepfake detection, real-time media analysis, and advanced document verification methods.
- **Cost-Effective System Implementation:** The system shall utilize open-source tools, frameworks, and libraries wherever possible in order to reduce development and operational costs while maintaining system performance and reliability.

Decision-Making Rules

Decision-making rules define how the system processes content and handles situations that may affect the accuracy or reliability of analysis results.

1. Content Preprocessing Decisions

When users upload videos or images that contain poor lighting conditions, low resolution, or excessive noise, the system applies preprocessing techniques such as normalization, enhancement, and noise reduction. These preprocessing steps improve the quality of input data before analysis. If facial features in a video or image are partially obscured, the system continues the analysis but adjusts the confidence score to reflect potential uncertainty. Similarly, when textual documents contain formatting inconsistencies or incomplete data, the system attempts to normalize the text before performing AI detection or similarity analysis.

2. Detection Confidence Evaluation

After processing content through AI models, the system evaluates the confidence score associated with the prediction. If the confidence score falls below a predefined reliability threshold, the system notifies the user that the result may be uncertain. Such cases are recorded within the system database and may later be used to improve the performance of the detection models through further training or evaluation.

3. Similarity Evaluation Rules

When academic documents are compared with existing records, similarity scores are calculated using text comparison algorithms. If the similarity score exceeds a predefined threshold, the system flags the document as potentially similar to existing submissions and presents the result to the user.

2.3.3 User Interface Prototypes

The user interface prototypes describe the visual layout and interaction structure of the Multi-modal AI Integrity Suite. The interface design focuses on simplicity, accessibility, and clarity so that users can easily interact with the system and understand the results of the analysis. The interface is structured into several major screens that guide the user through authentication, content analysis, document generation, and result interpretation.

Key UI Screens

Landing Page

The landing page introduces the purpose and functionality of the Multi-modal AI Integrity Suite. It provides navigation options that allow users to register, log in, and learn about the capabilities of the system.

- **User Authentication Pages**

Register Page: Allows new users to create an account by providing personal information such as name, email address, and password.

Login Page: Allows registered users to securely access the system using their login credentials.

Forgot Password Page: Allows users to request password reset instructions through their registered email.

Reset Password Page: Allows users to securely create a new password after verifying their identity.

- **Media Upload Interface:** This interface allows users to upload videos or images for authenticity detection. The page displays supported file formats, file size limits, and other upload requirements. Upload progress indicators and validation messages inform users whether their file meets the system requirements before analysis begins.

- **Academic Document Upload Interface:** This interface allows users to upload academic documents or project files for AI text detection and similarity analysis. Users can submit documents for originality verification and receive similarity scores indicating how closely the content matches existing records.
- **Detection Result Page:** This page displays the results of media or text analysis. The system presents classification outcomes such as Real or Fake media, AI-generated or human-written text, and similarity scores for academic documents. Each result includes a confidence score to indicate the reliability of the prediction.
- **Analysis Explanation View:** This page provides additional explanation regarding how the system reached a particular decision. The system may highlight detected visual inconsistencies in media files or structural patterns in text that influenced the AI model's prediction.
- **CV Generation Interface:** This page allows users to input personal information, education history, work experience, and skills. Based on this information, the system generates a professionally formatted CV that users can download or edit.
- **AI Chatbot Interface:** The chatbot interface enables users to ask questions regarding system usage, detection results, AI technologies, and content authenticity. The chatbot provides responses in a conversational format and helps users better understand the platform.
- **Analysis History Page:** This page displays previously analyzed content including media files and academic documents. Users can review past results, revisit similarity reports, and download previously generated analysis reports or CV documents.

2.3.4 State Chart Diagram

The state chart diagram illustrates the lifecycle of digital content within the Multi-modal AI Integrity Suite. It shows how different types of user-submitted content—videos, images, and academic documents—progress through the system from upload to final analysis, storage, or removal. This diagram models the dynamic behavior of the suite, ensuring that all content is handled securely and analyzed efficiently using AI models. When a user submits content through the system interface, it enters the Uploaded state. At this stage, the system receives the media file or document and prepares it for validation.

Validation

The system checks whether the uploaded file meets required conditions, including:

- Supported file formats (video, image, document)
- Acceptable file size limits
- Valid content structure

If these conditions are satisfied, the content transitions to the Validated state, confirming it is ready for further processing.

Processing

In the Processing state, the system prepares content for analysis:

- Videos: Frames are extracted, and facial or relevant visual regions are identified.
- Images: Key visual features are extracted for manipulation detection.
- Academic Documents: Text is preprocessed, normalized, and tokenized for natural language analysis.

Analysis

During the Analysis state, AI models verify the authenticity of the content:

- Computer Vision Models assess videos and images for deepfake artifacts or manipulations.
- NLP and Transformer Models analyze academic text to detect AI-generated content and compute similarity with previously stored submissions.

Classification

After analysis, content enters the Classified state. Here, the system generates final results:

- Media Files: Labeled as authentic or manipulated with a confidence score.
- Text Documents: Determined as human-written or AI-generated, with similarity scores if applicable.

Storage

Analysis results and associated metadata move to the Stored state, where they are securely saved in the system database. Users can later access this information through the platform's history or dashboard.

Deletion

Finally, users may choose to remove content or results. When this occurs, the system enters the Deleted state, permanently removing both the content and its associated results.

Key States

- **Uploaded:** Content is successfully uploaded to the system.
- **Validated:** File format, size, and integrity are confirmed.
- **Processing:** Preprocessing of videos, images, or text is performed.
- **Analysis:** AI models detect deepfakes, AI-generated text, or project similarities.
- **Classified:** Analysis results, including authenticity labels and confidence scores, are generated
- **Stored:** Results and metadata are saved for future reference.
- **Deleted:** Content and results are permanently removed from the system.

1.State chat Diagram for Authentication State Diagram (Login / Register)

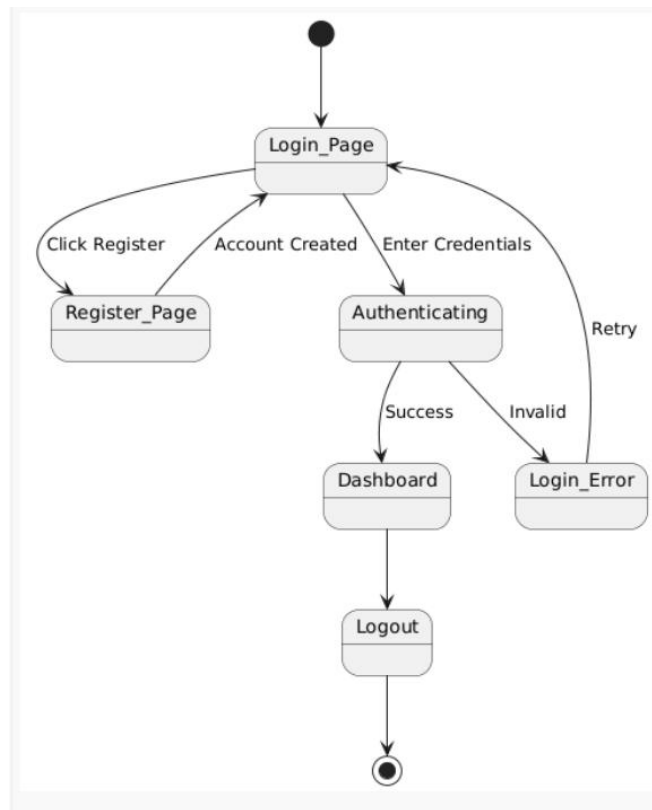


Figure 2: State chat Diagram for Authentication State Diagram (Login / Register)

2. State Chart Diagram for Video/Image Detection

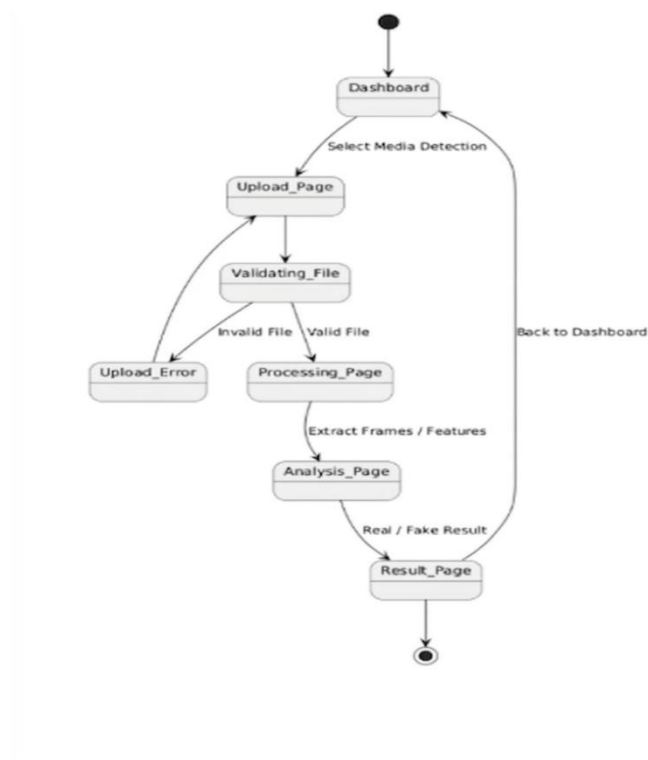


Figure 3: State Chart Diagram for Video/Image Detection

3. State chat diagram for Ai Text Detection

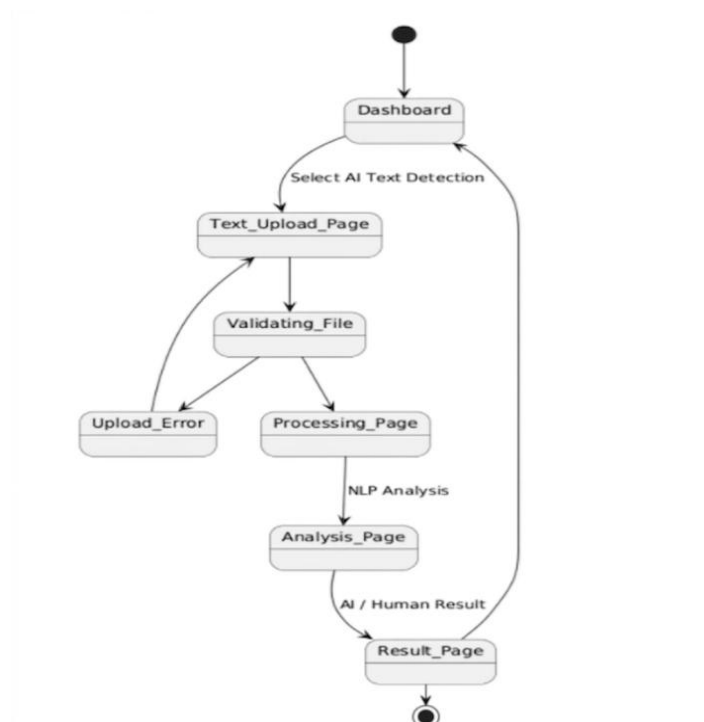


Figure 4: State chat diagram for Ai Text Detection

4.State chat diagram for Academic Similarity Checking

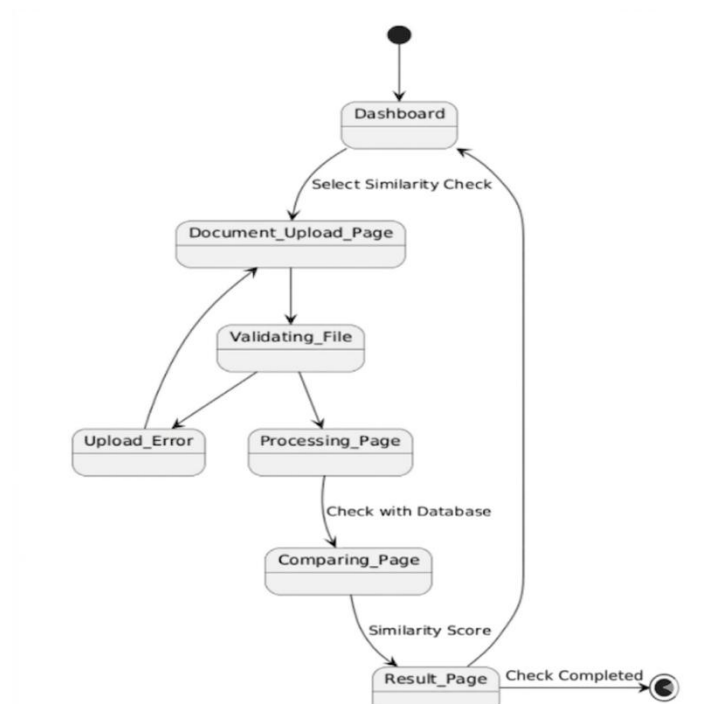


Figure 5:State chat diagram for Academic Similarity Checking

5.State chat diagram for Cv generation

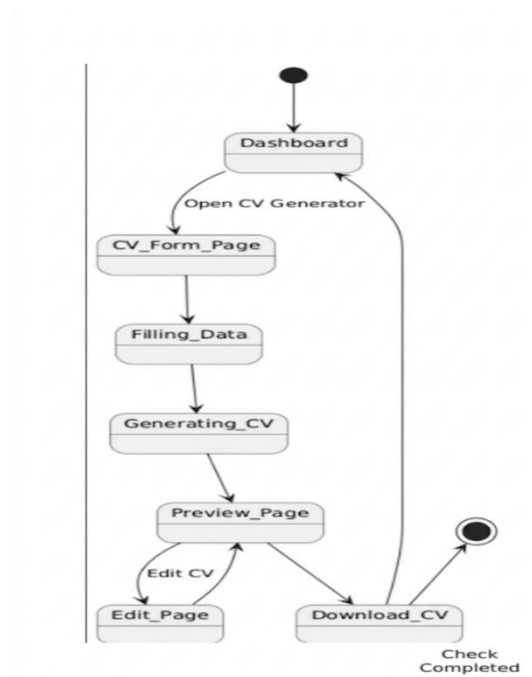


Figure 6:State chat diagram for CV generator

6.State chat diagram for Chat Bot interaction

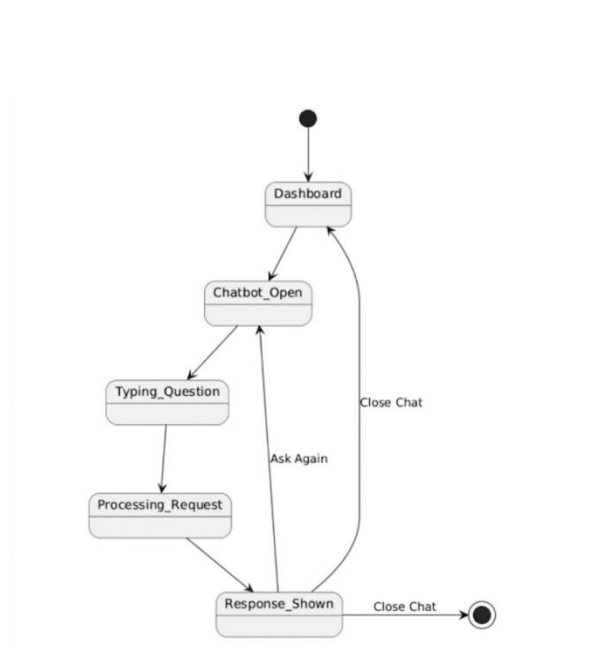


Figure 7:State chat diagram for Chat Bot interaction

Details of the State Chart Diagram

The State Chart Diagram represents the dynamic behavior of the Multi-modal AI Integrity Suite by illustrating how the system transitions between different states during user interaction. It shows how users move through various system interfaces such as the login page, dashboard, upload pages, processing stages, and result viewing screens. The diagram helps in understanding how the system responds to user actions and how different processes such as validation, analysis, and result generation are executed in sequence.

The system begins from an initial state where the user accesses the platform through the login page. After successful authentication, the user is directed to the dashboard, which serves as the central hub for accessing all functionalities. From the dashboard, the user can navigate to different modules such as media detection, text analysis, similarity checking, CV generation, and chatbot interaction. Each module follows a structured flow that includes data input, validation, processing, analysis, and result presentation.

During content-based operations, such as media or document analysis, the system first validates the uploaded file to ensure it meets the required format and size constraints. If the file is invalid, the system transitions to an error state and prompts the user to re-upload. If valid, the system proceeds to processing and analysis states where artificial intelligence models perform tasks such as deepfake detection, AI text identification, or similarity comparison. Once analysis is complete, the system transitions to a result state where outputs such as classification labels, confidence scores, or similarity percentages are displayed to the user.

In addition to analysis functionalities, the system also supports CV generation, where users input their personal and professional information, preview the generated document, optionally edit it, and finally download the CV. The chatbot interaction state allows users to seek assistance, ask questions, and better understand system results.

The diagram also includes termination states such as logout or content deletion, representing the end of a session or workflow. Overall, the State Chart Diagram provides a clear visualization of system behavior, ensuring that all processes are logically connected and user interactions are efficiently managed.

- The diagram models the dynamic behavior of the system during user interaction

- It starts from the login page and transitions to the dashboard after authentication
- The dashboard acts as the central state for accessing all system features
- Each functionality (media detection, text analysis, CV generation) follows a structured flow
- Validation state ensures uploaded content meets system requirements
- Error states handle invalid inputs and allow users to retry
- Processing and analysis states execute AI-based operations
- Result state displays outputs such as:
 - Real/Fake classification
 - AI-generated text detection
 - Similarity scores
- The system includes interactive states such as chatbot assistance
- Users can return to the dashboard or terminate the session through logout
- The diagram ensures clear, logical, and efficient system flow

2.3.5 Activity Diagram

The activity diagram illustrates the step-by-step workflow of the Multi-modal AI Integrity Suite, showing how users and the system interact during content submission, AI analysis, and result viewing. It captures the complete flow for videos, images, academic documents, and CV generation.

Main Activity Flow

1. Start: The user accesses the system through the web dashboard.
2. User Authentication: The user registers or logs in to access the platform.
3. Content Submission: The user uploads a video, image, or academic document, or enters information for CV generation.
4. Validation: The system checks content for:
 - Supported file types and formats
 - Acceptable file sizes
 - Data integrity and proper structure

5. Processing: The system prepares the content for analysis:
 - Videos: Frame extraction and region detection
 - Images: Visual feature extraction
 - Academic Documents: Text preprocessing and normalization
6. AI Analysis: The system performs integrity verification:
 - Video/Image Deepfake Detection: Using CNN and computer vision models
 - Academic Text Analysis: Using NLP/Transformer models to detect AI-generated content and calculate similarity scores
 - CV Generation: Automatic compilation of professional CV based on user input
7. Result Classification: The system generates outputs:
 - Media Files: Authentic or manipulated, with confidence score
 - Text Documents: Human-written or AI-generated, with similarity score
 - Generated CV: Professional document ready for download
8. Result Storage: Results and metadata are saved securely in the system database for future access.
9. User Interaction (Optional): The user may interact with a help chatbot to understand results or ask questions about content verification.
10. End: The user logs out or exits the system.

Activity Diagram Description

The activity diagram illustrates the step-by-step workflow of the AI-Based Deepfake and Synthetic Video Detection System, showing how users and the system interact during video upload, deepfake detection, and result viewing.

Main Activity Flow

- Start : The user accesses the system.
- User Authentication : The user registers or logs in to the system.
- Upload Video : The user submits a video for analysis.
- Validate Video : The system checks the video format, size, and duration.
- Detect Deepfake : The system analyzes the video using AI models.
- View Result : The user views the classification result and confidence score.

- Interact with Chatbot (Optional) : The user asks questions related to the detection process or results.
- End : The user logs out or exits the system.

Activity Diagram for the video detection

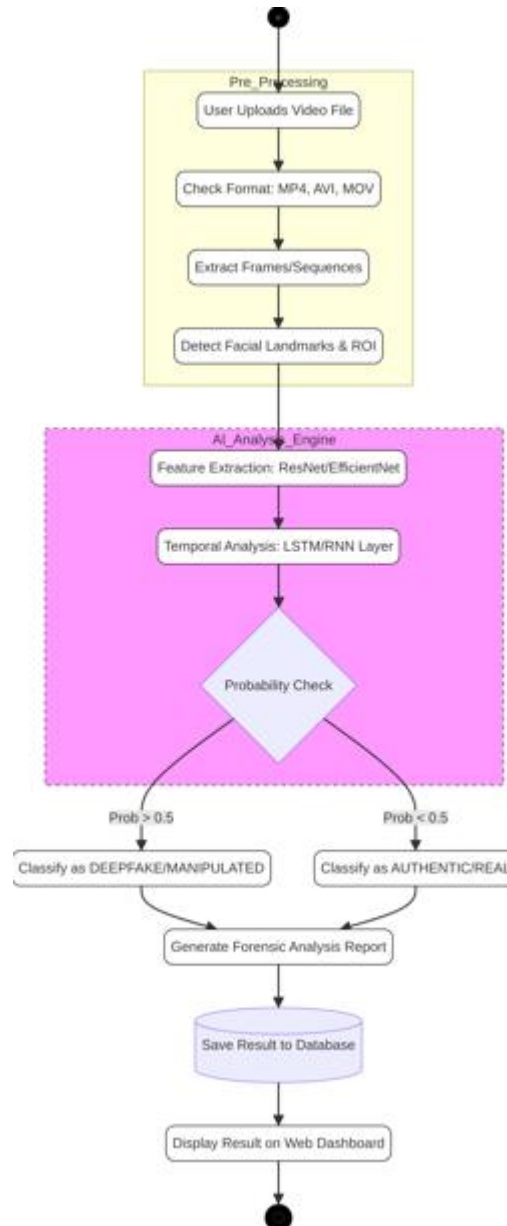


Figure 8:Activity Diagram for the video detection

Activity Diagram Description

The activity diagram describes the logic for detecting AI-generated or "Photoshopped" images. It focuses on pixel-level forensics, Error Level Analysis (ELA), and metadata verification to ensure image authenticity.

Main Activity Flow

- Start : The user navigates to the image integrity section.
- User Authentication : The user logs in to access advanced forensic tools.
- Upload Image : The user uploads a suspicious image (JPG/PNG).
- Forensic Scanning : The system performs Error Level Analysis (ELA) to find compression artifacts and checks EXIF metadata.
- AI Feature Extraction : A Vision Transformer (ViT) or CNN analyzes the image for synthetic textures common in AI generators (like DALL-E or Midjourney).
- Classification : The system determines if the image is "Authentic," "Manually Edited," or "AI-Generated."
- View Analysis Report : The user reviews the visual heatmaps and the final verdict.
- End : The user downloads the integrity certificate or exits.

Activity Diagram for the Image detection

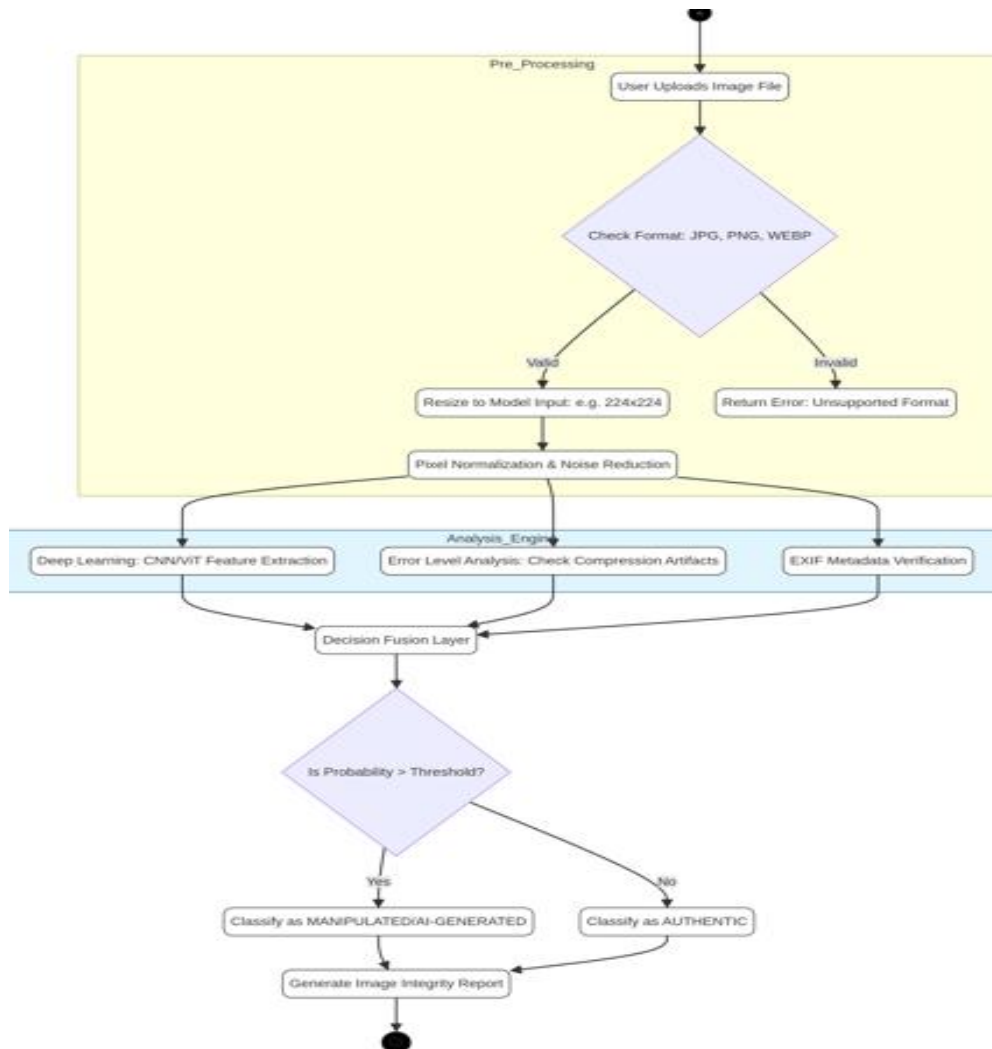


Figure 9;Activity Diagram for the video detection

Activity Diagram Description

The activity diagram illustrates the process of verifying academic originality. It shows how student projects are compared against the Bahir Dar University (BDU) internal database and scanned for AI-generated text patterns.

Main Activity Flow

- Start : The user (Student or Admin) accesses the Plagiarism Checker.
- User Authentication : The user logs in to the academic integrity portal.
- Upload Document : The user submits a research paper or project file.
- Text Extraction & Cleaning : The system parses the document text and removes bibliography/references from the count.

- Similarity Search : The system compares the text against the BDU Project Database and external academic sources.
- AI Text Detection : The Transformer model scans the text for linguistic patterns typical of LLMs (like ChatGPT).
- Generate Integrity Report : The system produces a percentage score for both similarity (plagiarism) and AI-generated content.
- End : The Administrator approves or flags the submission based on the result.

Activity Diagram for the Plagiarism Checker

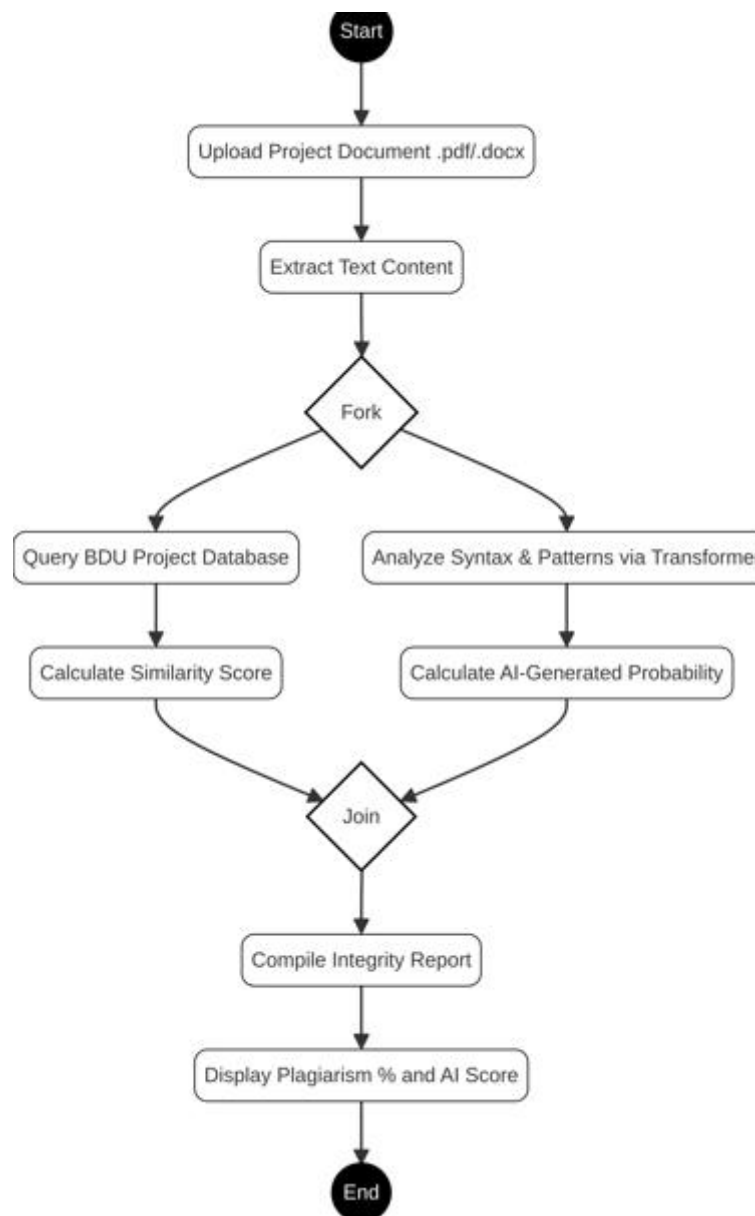


Figure 10:Activity Diagram for the Plagiarism Checker

Activity Diagram Description

The activity diagram outlines the automated workflow for creating professional career documents. It shows how user input is transformed into an optimized CV using AI-driven content enhancement and professional formatting templates.

Main Activity Flow

- **Start** : The user selects the CV Generation tool.
- **User Authentication** : The user logs in to save their professional profile.
- **Input Data** : The user enters personal details, education, skills, and work experience.
- **Select Template** : The user chooses a professional layout from the system library.
- **AI Content Optimization** : The AI engine refines the user's descriptions, improves grammar, and suggests industry-specific keywords for ATS optimization.
- **Preview & Edit** : The user reviews the generated CV and makes manual adjustments if necessary.
- **Generate & Download** : The system compiles the data into a PDF or Word document.
- **End** : The user saves the CV to their profile and exits.

Activity Diagram for the CV Generator

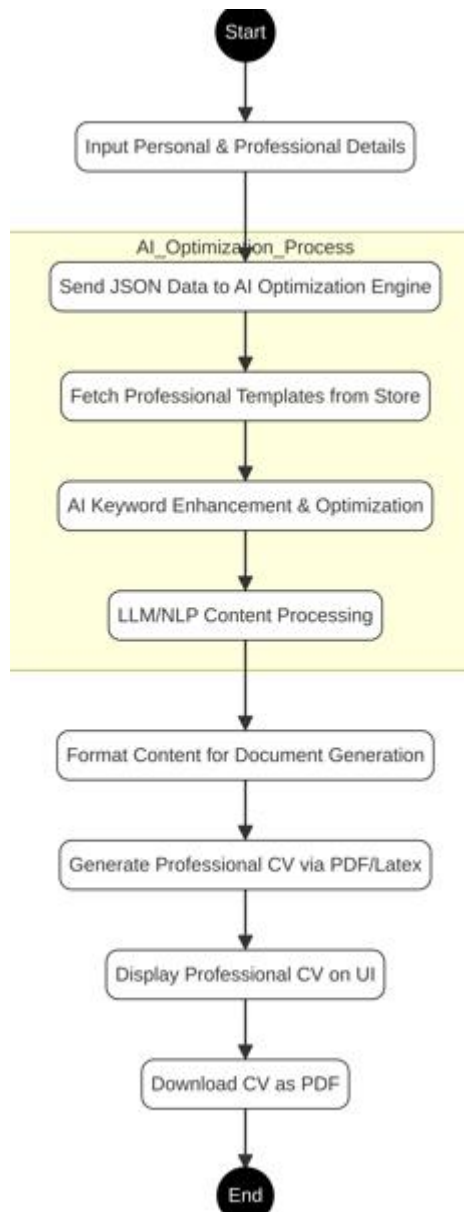


Figure 11: Activity Diagram for the CV Generator

2.3.5 Sequence Diagram

The sequence diagram illustrates the interaction between users and system components over time in the Multi-modal AI Integrity Suite, showing how requests are sent, processed, and responded to for different content types and functionalities.

Actors and Components

- User: Submits videos, images, academic documents, or CV information.

- Web Dashboard: Provides the interface for content upload, result viewing, and interaction.
- Authentication Module: Handles user registration and login.
- Validation Module: Checks file formats, size limits, and data integrity.
- Processing Module: Prepares content for AI analysis (frame extraction, feature extraction, text preprocessing)
- AI Analysis Module: Performs authenticity verification using:
 - Computer Vision Models for videos/images
 - NLP/Transformer Models for text
 - CV Generation Engine for professional document creation
- Database: Stores analysis results, similarity scores, and generated CVs.
- Chatbot (Optional): Provides user guidance and explanations for results.

Sequence Flow

1. User Accesses System: Initiates session via the web dashboard.
2. Authentication: User registers or logs in. Authentication module confirms credentials.
3. Content Submission: User uploads a video, image, academic document, or enters CV data.
4. Validation: System verifies file type, format, size, and content integrity.
5. Processing: Content is preprocessed:
 - Videos: Frames extracted, regions detected
 - Images: Key features extracted
 - Documents: Text normalized and tokenized
6. AI Analysis: The system analyzes the processed content:
 - Deepfake detection for videos/images
 - AI-generated text detection and similarity analysis for documents
 - CV generation based on user input
7. Result Generation: System classifies content and/or produces a CV.
8. Storage: Analysis results, similarity scores, and generated CVs are saved in the database.
9. User Notification: Results are sent back to the user interface for viewing.
10. Optional Chatbot Interaction: User may ask questions related to results or system workflow.

11. Session End: User logs out or closes the system interface.

1. Sequence diagram for Authentication(Login and signup)

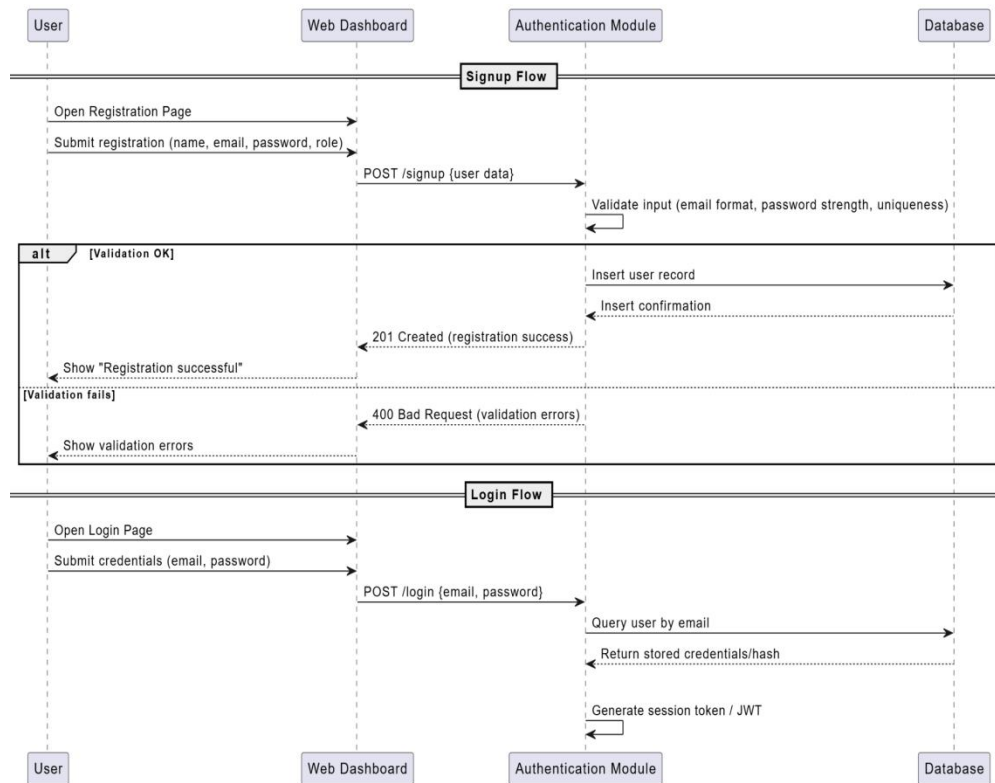


Figure 12: .Sequence diagram for Authentication(Login and signup)

2. Sequence Diagram for Video Deepfake Detection

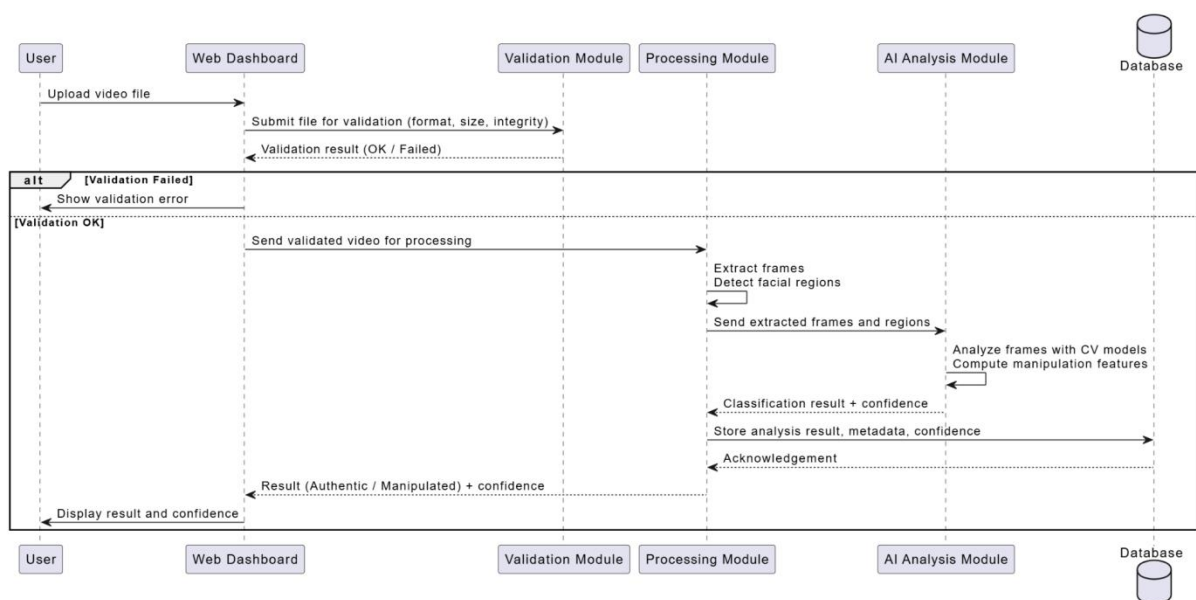


Figure 13: Sequence Diagram for Video Deepfake Detection

3. Sequence Diagram for Image Deepfake Detection

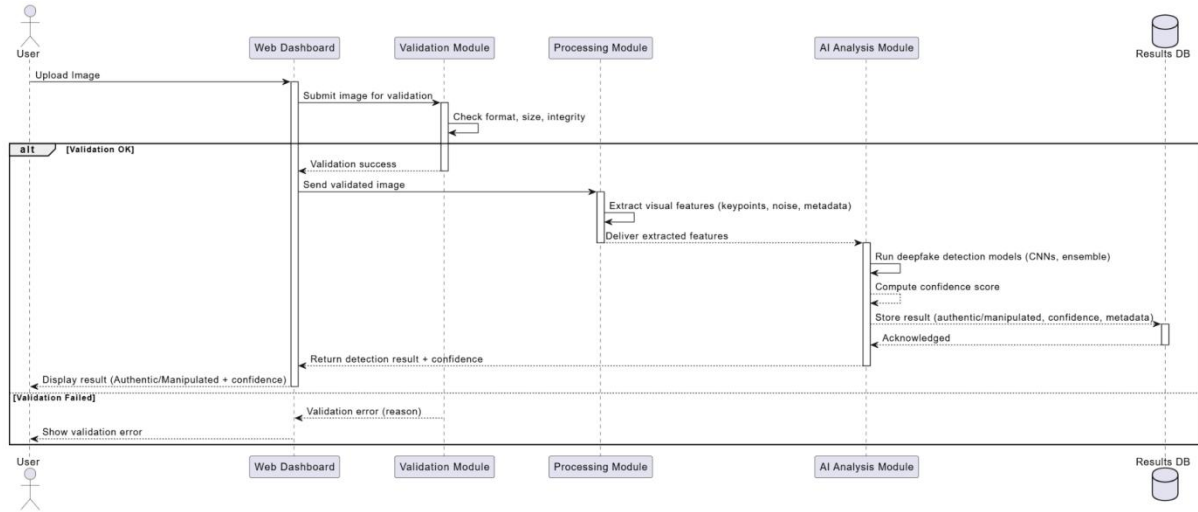


Figure 14: Sequence Diagram for Image Deepfake Detection

4. Sequence diagram for Text AI Detection

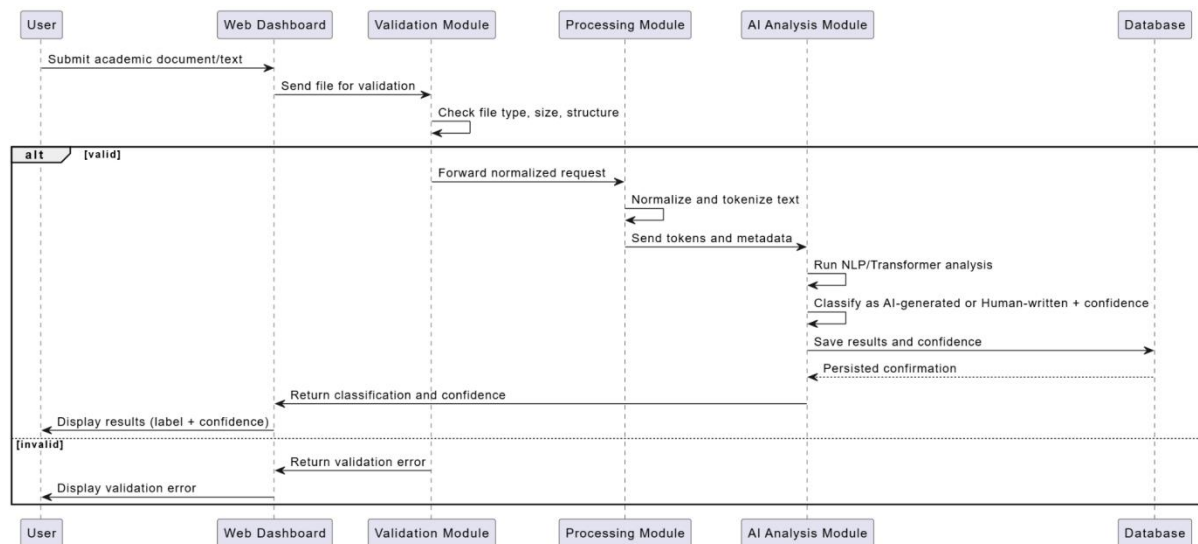


Figure 15: Sequence diagram for Text AI Detection

5. Sequence diagram for Project Similarity Checking

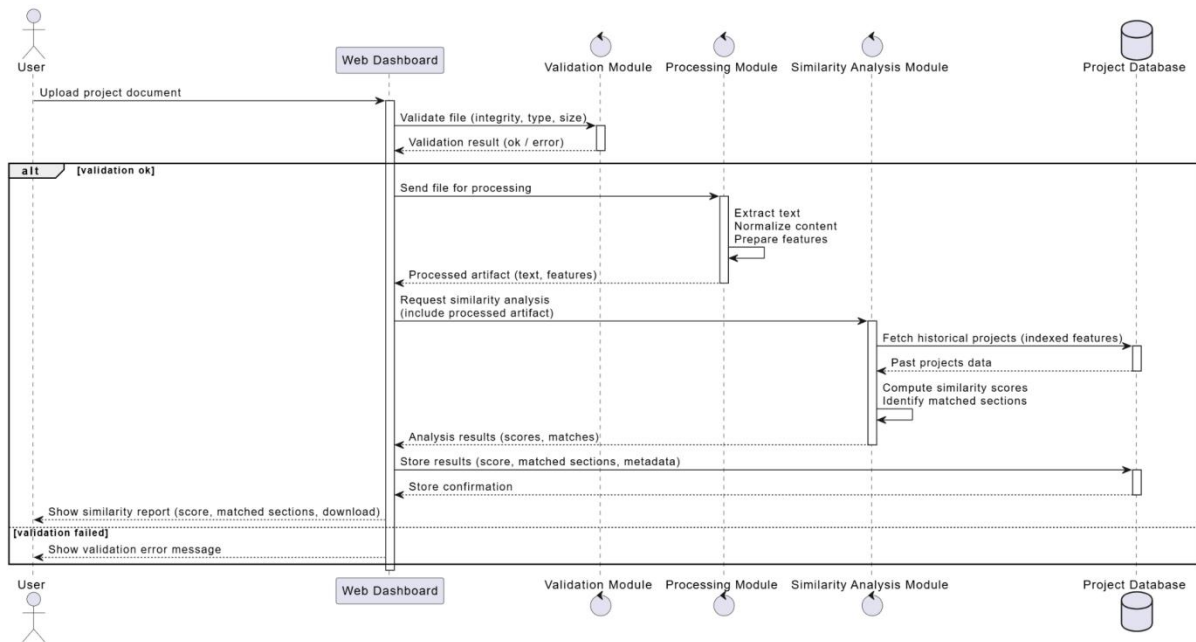


Figure 16: Sequence diagram for Project Similarity Checking

6. Sequence Diagram for Automated CV Generation

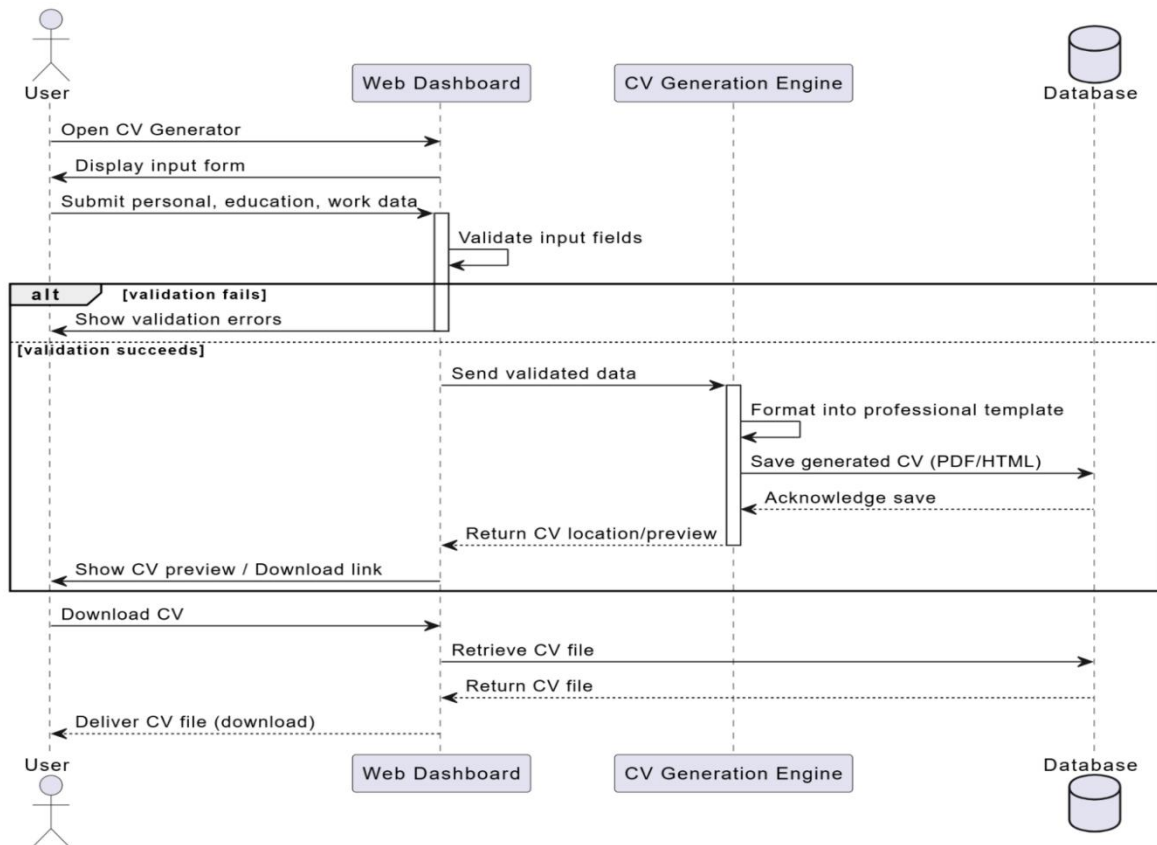


Figure 17: Sequence Diagram for Automated CV Generation

7. Sequence Diagram For Chatbot Interaction

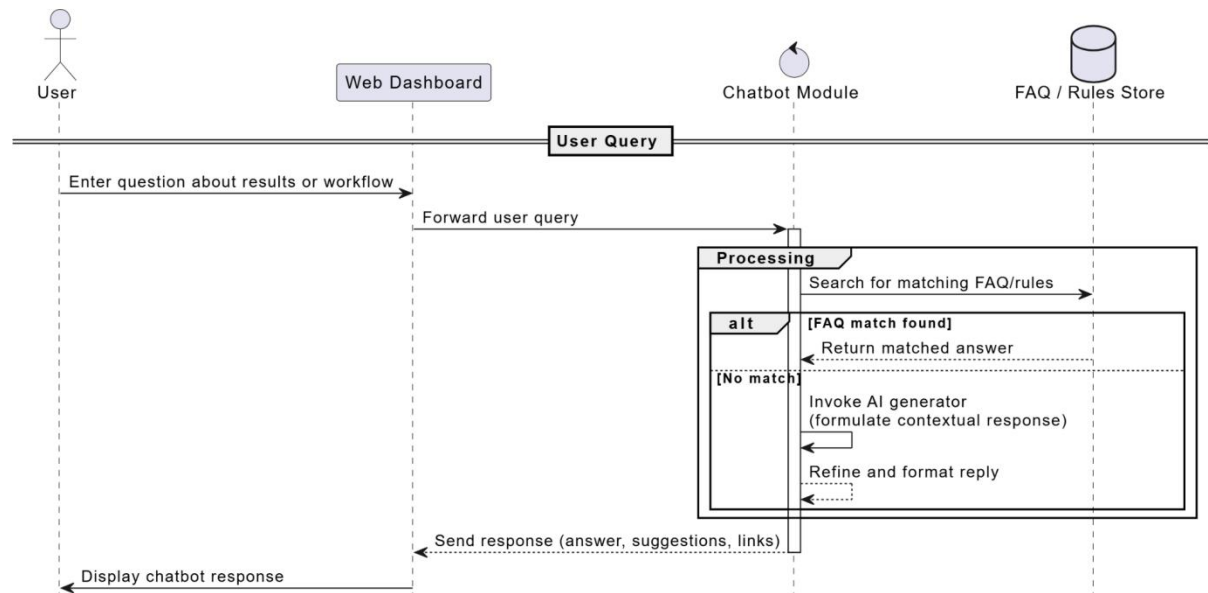


Figure 18: Sequence Diagram For Chatbot Interaction

2.3.8 Analysis Class Model

The Analysis Class Model provides a high-level, conceptual representation of the Multi-modal AI Integrity Suite, outlining its primary entities, their attributes, behaviors, and interrelationships. This model is crucial for understanding the system's static structure and how its various components collaborate to achieve the overarching goals of digital authenticity, academic integrity, and professional development. It abstracts away implementation details, focusing instead on the core responsibilities and interactions of key classes such as User, MediaContent (for videos and images), TextDocument, AcademicProject, CVProfile, DeepfakeDetector, AITextDetector, SimilarityChecker, CVGenerator, and ResultManager. This foundational model guides the subsequent design phases, ensuring a cohesive and comprehensive system architecture that addresses all specified functional requirements.

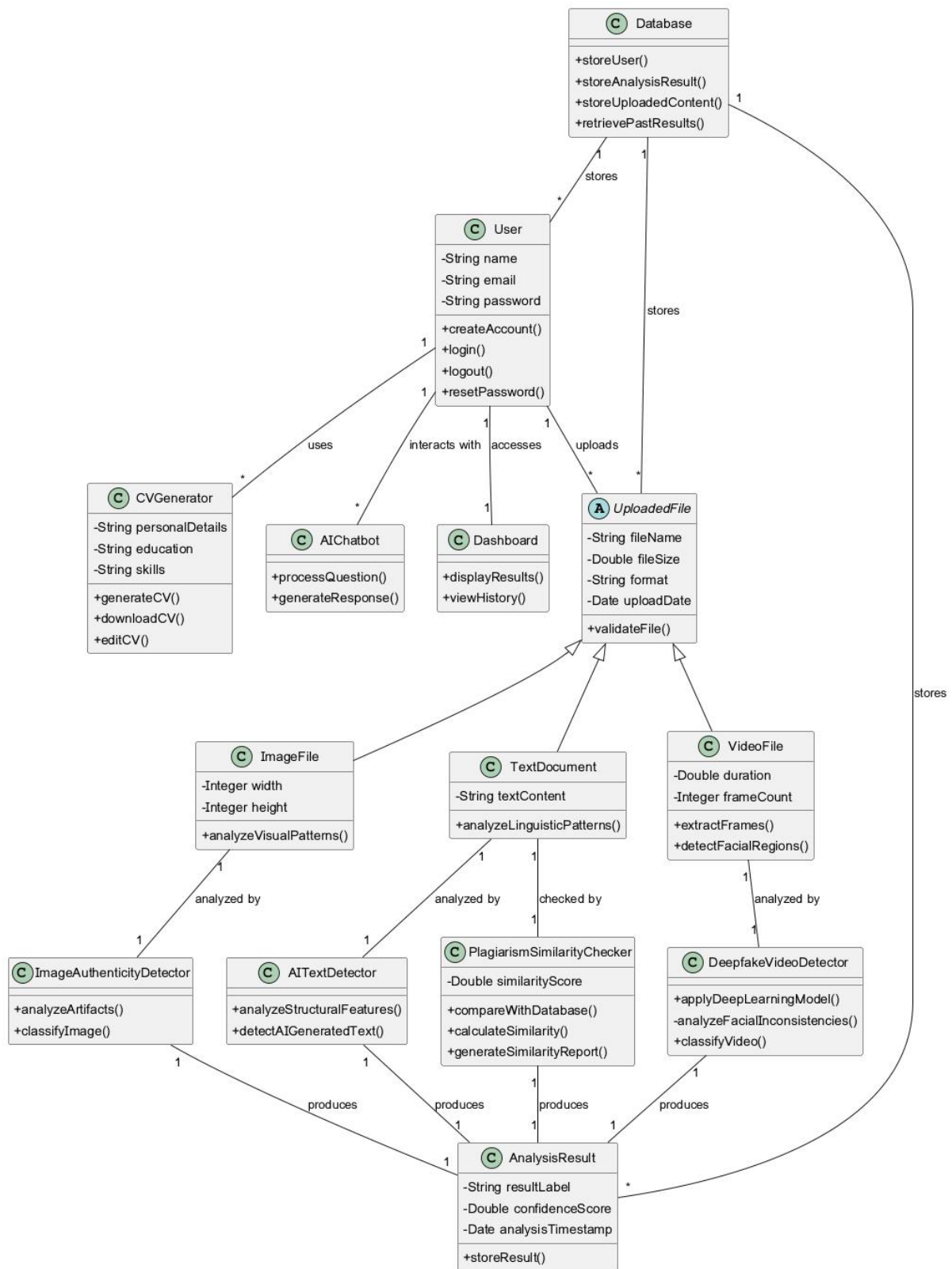


Figure 19: Analysis Class Model

2.4 Non-Functional Requirements

Non-functional requirements (NFRs) are paramount for defining the quality attributes and operational constraints of the Multi-modal AI Integrity Suite, ensuring its robustness, efficiency, and user satisfaction across all its diverse functionalities. These requirements dictate the system's performance, usability, security, reliability, scalability, maintainability, compatibility, interoperability, and adherence to legal and ethical standards.

1. Performance

Performance requirements are critical for handling various data types efficiently. This includes processing time for detecting deepfakes in videos and images, identifying AI-generated text, performing similarity checks on academic projects, and generating CVs. All operations must be optimized for rapid analysis and response. Model efficiency is essential, requiring deep learning and NLP models to deliver high accuracy with reasonable inference times across all detection tasks. Furthermore, resource utilization must be optimized to minimize CPU/GPU usage, memory consumption, and processing overhead, especially when dealing with large media files or extensive text corpora.

2. Usability

Usability focuses on providing an intuitive and accessible experience for all users, from students to educators. An intuitive interface is crucial, allowing users to easily upload various content types (videos, images, text, project files) and generate CVs with minimal technical knowledge. The UI design must clearly display upload status, detection results (e.g., deepfake probability, AI-generated text score, similarity percentage), and confidence scores, using clear visual indicators. User guidance should be integrated throughout the platform, offering clear instructions and informative messages for each module, including CV generation prompts and similarity report interpretations.

3. Security

Security requirements are vital for protecting sensitive user data and maintaining the integrity of analysis results. Data protection is ensured through secure HTTPS transmission for all uploads and downloads. User-submitted content (videos, images, academic projects, CV data) will be handled with strict access controls and encrypted storage, with clear policies on data

retention. Access control mandates robust login and registration mechanisms, with role-based access for different user types (e.g., students, instructors, administrators). Input validation is crucial to prevent malicious or unsupported files from compromising the system. Finally, privacy assurance ensures that all data handling adheres to ethical guidelines and relevant data protection regulations (e.g., GDPR, FERPA).

4. Reliability

Reliability ensures the system consistently performs its intended functions across all modules without failure. This involves accuracy and consistency in deepfake detection, AI text identification, and similarity checking, producing dependable results for similar inputs. Fault tolerance is necessary to handle unexpected errors gracefully across all functionalities, preventing system crashes. Comprehensive error handling mechanisms, including detailed logging for debugging and user-friendly error messages, are vital for system robustness and user trust.

5. Scalability

Scalability addresses the system's ability to handle increasing workloads and growth across its diverse functionalities. System scalability supports expansion for larger datasets, more users, and a growing volume of content uploads (videos, images, text, projects). Model scalability allows for the seamless integration of improved or additional deep learning and NLP models for deepfake detection, AI text detection, and similarity analysis without requiring major architectural redesigns. Deployment scalability facilitates cloud deployment to accommodate higher computational demands and ensure continuous availability for all services.

6. Maintainability

Maintainability focuses on the ease with which the system can be modified, updated, and repaired across its various modules. A modular design is essential, simplifying updates and maintenance efforts for each distinct functionality (deepfake, AI text, similarity, CV). Code readability is promoted through adherence to coding standards and comprehensive documentation for all components. Model update support enables the retraining or replacement of detection models with new datasets, ensuring the system remains current and

effective against evolving deepfake techniques, AI text generation methods, and academic plagiarism strategies.

7. Compatibility

Compatibility ensures the system functions across various user environments and integrates with necessary tools. Platform compatibility includes support for major operating systems (Windows, Linux, macOS). Browser compatibility ensures seamless operation across popular web browsers (Chrome, Firefox, Edge). File format compatibility extends to supporting common video (MP4, AVI), image (JPEG, PNG), and document (PDF, DOCX, TXT) formats for analysis, as well as standard CV output formats.

8. Interoperability

Interoperability allows the system to interact effectively with other systems and components. Framework integration ensures compatibility with key deep learning (TensorFlow, PyTorch), computer vision (OpenCV), and NLP (Hugging Face, NLTK) frameworks. API communication facilitates interaction between all system components (frontend, backend, detection modules, CV generator) via RESTful APIs, promoting a loosely coupled and extensible architecture.

9. Legal & Ethical Compliance

This category ensures the system operates within legal and ethical boundaries, particularly given its sensitive applications in academic integrity and personal data. Ethical AI principles dictate that the system is used solely for ethical purposes, such as media verification, academic integrity checks, and career assistance. Data privacy compliance adheres to applicable data protection regulations (e.g., GDPR, FERPA) for all user-submitted content and personal information. Content responsibility ensures the system is not used for creating or manipulating media, upholding its integrity and preventing misuse.

10. Extensibility & Future Enhancements

Extensibility and future enhancements outline the system's capacity for growth and adaptation to emerging challenges. Feature extension includes potential future capabilities like real-time detection, advanced multi-modal analysis combining video, image, and text cues, and integration with learning management systems. Technology adaptability allows for

the adoption of new AI models and detection techniques with minimal system changes. Finally, open-source contribution encourages community-driven enhancements, enabling continuous improvement and broader adoption of the suite.

2.5 System Requirements

To ensure optimal performance and functionality across its diverse modules, the Multi-modal AI Integrity Suite necessitates specific hardware and software configurations.

2.5.1 Hardware Requirements

For the efficient operation of the Multi-modal AI Integrity Suite, certain hardware specifications are recommended to handle the computational demands of deepfake detection, AI text analysis, similarity checking, and CV generation. A minimum of a quad-core processor (e.g., Intel Core i5 or equivalent) is required, with an Intel Core i7/i9 or AMD Ryzen 7/9 (or equivalent server-grade CPUs) preferred for intensive deep learning tasks and parallel processing of multiple requests. Regarding memory, at least 16 GB of RAM is necessary, with 32 GB or more recommended for smooth execution during model training, large-scale data processing, and concurrent user sessions. Storage-wise, a minimum of 500 GB of available SSD storage is required for faster read/write operations, with 1 TB or more recommended for storing diverse datasets (video, image, text), extracted features, and trained models. An NVIDIA GPU with CUDA support (e.g., RTX 3060 or higher) is highly recommended to significantly accelerate deep learning model training and inference for deepfake and AI text detection; while the system can function without it, performance will be substantially reduced. A stable, high-bandwidth internet connection is essential for accessing public datasets, installing libraries, updating components, and facilitating user uploads/downloads. Standard peripherals, including a 1080p monitor, keyboard, and mouse, are also required for development and user interaction.

2.5.2 Software Requirements

The Multi-modal AI Integrity Suite is designed for platform independence, supporting development and deployment across Windows, Linux, and macOS. Python (version 3.9 or higher) is the primary programming language for deep learning, computer vision, natural language processing, and data analysis components. JavaScript/TypeScript supports both backend and frontend development. The backend is implemented using Node.js (latest LTS

version) with Express.js or FastAPI (Python) for robust API requests and server-side logic, handling diverse data types. The frontend utilizes Next.js or React for a dynamic and responsive web interface, with TypeScript for type safety and maintainability, and Tailwind CSS or Material-UI for responsive UI styling. Core deep learning and computer vision functionalities are powered by TensorFlow/PyTorch for model training/inference, OpenCV for video/image processing, and Dlib/MTCNN for facial detection. Natural Language Processing for AI text detection and similarity checking will leverage libraries like Hugging Face Transformers, NLTK, and SpaCy. Advanced neural networks (e.g., XceptionNet, EfficientNet, BERT, GPT-variants) will be used for feature extraction and classification. For data storage, MongoDB (NoSQL) is employed for flexible document storage (user data, results, logs) and PostgreSQL (SQL) for structured data (academic project metadata, user profiles). Development tools include Visual Studio Code/PyCharm, Kaggle/Hugging Face Datasets for datasets, and Git/GitHub for version control. Frontend testing is primarily on Google Chrome, with Firefox and Edge for cross-browser compatibility. API testing and debugging are supported by Postman/Insomnia and Python/Node.js logging tools.

2.6.1 Conceptual Modeling: Class Diagram

The Class Diagram for the Multi-modal AI Integrity Suite visually represents the system's conceptual model, illustrating the classes, their attributes, operations, and the relationships between them. This diagram is a cornerstone of object-oriented design, offering a comprehensive, high-level view of the system's structure and how different components interact to fulfill the diverse functional requirements. It expands upon the CRC (Class, Responsibility, Collaboration) analysis by providing a standardized graphical notation that clarifies the system's architecture and facilitates communication among development teams. The class diagram will depict key entities such as User, MediaContent (parent class for Video and Image), TextDocument, AcademicProject, CVProfile, DeepfakeDetector, AITextDetector, SimilarityChecker, CVGenerator, and ResultManager, along with their associations, aggregations, and compositions. This detailed visual model lays the groundwork for the precise design and implementation of the system's software components, ensuring all aspects of deepfake detection, AI text detection, similarity checking, and CV generation are robustly integrated.

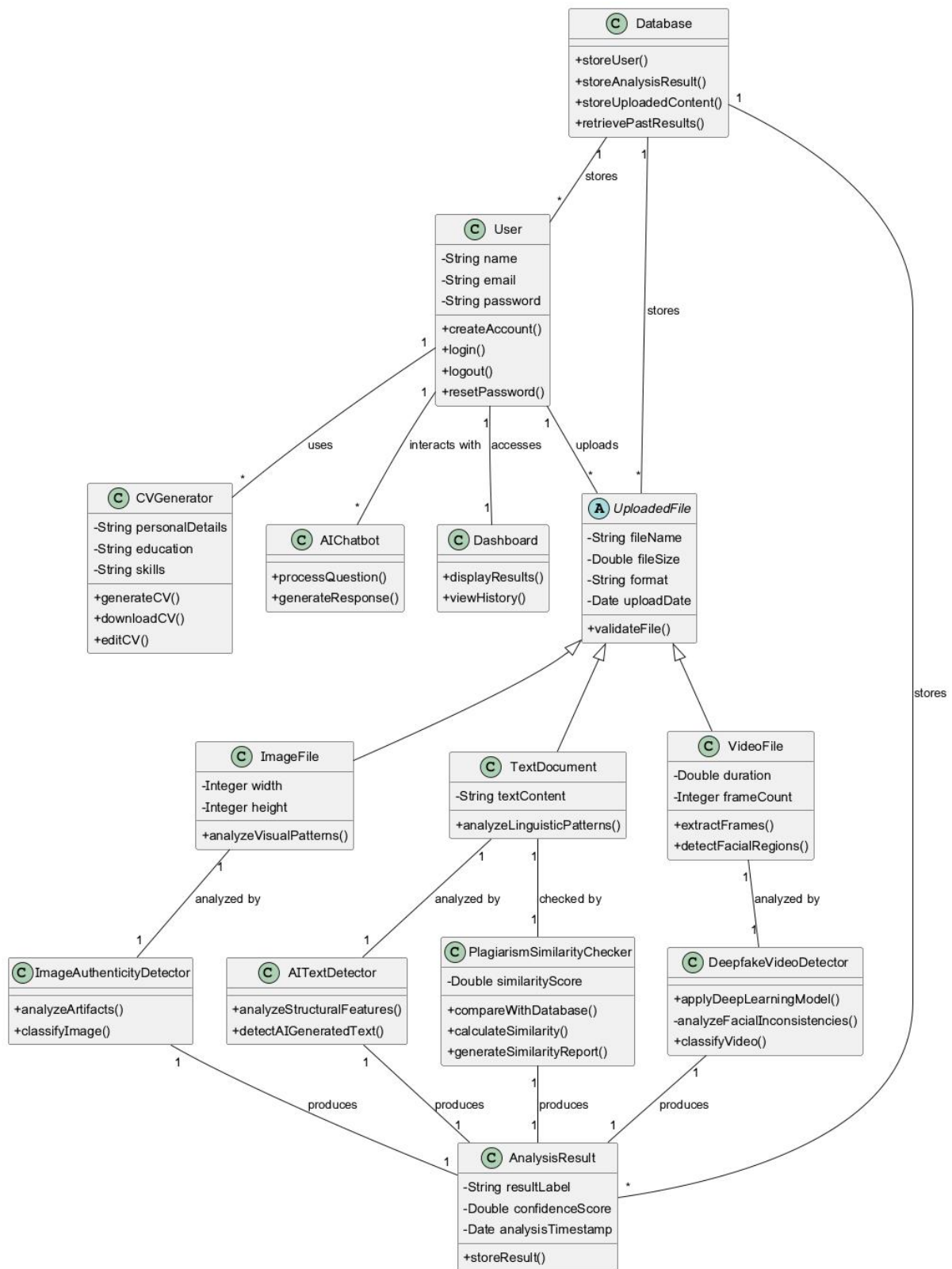


Figure 20: Conceptual Modeling: Class Diagram

2.6.2 Identifying Change Cases

Anticipating future enhancements and adaptations is crucial for the long-term viability and relevance of the Multi-modal AI Integrity Suite, given the rapidly evolving landscape of AI technologies and digital threats. Several potential change cases have been identified to ensure the system remains scalable, adaptable, and future-proof across all its functionalities:

1. **Advanced Deepfake Modalities:** Extend deepfake detection to include audio deepfakes (voice manipulation, synthetic speech) and multi-modal deepfakes (combining video, audio, and text inconsistencies). This is a high-impact change requiring new processing modules and integrated classifiers.

2. **Enhanced AI Text Detection Models:** Continuously update and integrate newer, more sophisticated NLP models (e.g., larger transformer models, specialized adversarial detection models) to combat increasingly advanced AI text generation techniques. This is a high-likelihood, medium-impact change.

3. **Cross-Lingual Support:** Implement capabilities for AI text detection and similarity checking across multiple languages, requiring integration of multilingual NLP models and translation services. This is a medium-likelihood, high-impact change.

4. **Real-Time Processing:** Develop real-time or near real-time processing capabilities for video streams and live text inputs, particularly for deepfake detection and AI text identification in dynamic environments. This is a low-likelihood but high-impact change, demanding significant optimization and infrastructure.

5. **Integration with Learning Management Systems (LMS):** Provide seamless API integration with popular LMS platforms (e.g., Moodle, Canvas, Blackboard) for automated submission of academic projects for similarity and AI text checks. This is a medium-likelihood, high-impact change.

6. **Interactive CV Customization:** Enhance the CV generation module with more advanced customization options, AI-driven content suggestions based on job descriptions, and integration with professional networking platforms. This is a medium-likelihood, medium-impact change.

7. User Feedback and Model Retraining Loop: Implement a system for collecting user feedback on detection results to continuously retrain and improve the accuracy of deepfake and AI text detection models. This is a high-likelihood, medium-impact change.

8. Blockchain for Authenticity Verification: Explore the use of blockchain technology to create immutable records of content authenticity checks, providing verifiable proof of originality for academic work or media. This is a low-likelihood, high-impact change.

9. Expanded Media Format Support: Continuously expand support for new and emerging video, image, and document formats as they become prevalent. This is a high-likelihood, low-impact change.

10. Containerization and Microservices Architecture: Transition to a fully containerized (e.g., Docker, Kubernetes) and microservices-based architecture to enhance deployment flexibility, scalability, and independent development of each module. This is a medium-likelihood, high-impact architectural change.

Chapter 3: System Design

This chapter presents the purpose and design goals of the proposed Multi-modal AI Integrity Suite: Deepfake Detection, Content Authentication, and Intelligent Document Generation System. The design of the system is guided by several important quality attributes including performance, reliability, security, scalability, usability, and maintainability. These attributes influence the system architecture, module organization, and technology selection to ensure that the system operates efficiently and meets the functional and practical requirements of users.

The system integrates multiple artificial intelligence techniques to analyze different types of digital content including videos, images, and academic text, while also providing an automated CV generation service. By combining these capabilities within a unified platform, the system aims to enhance digital authenticity verification and support academic integrity.

Purpose of the System

The purpose of the Multi-modal AI Integrity Suite is to provide an integrated platform capable of detecting manipulated digital media, identifying AI-generated academic content, verifying student project originality, and generating professional curriculum vitae automatically.

With the rapid growth of generative artificial intelligence technologies, it has become increasingly difficult to verify the authenticity of digital media and written content. Manipulated videos, synthetic images, and AI-generated text pose significant challenges to academic institutions, organizations, and individuals. This system addresses these challenges by leveraging deep learning, computer vision, and natural language processing techniques to analyze different forms of digital content within a unified framework.

The system enables users to upload videos, images, or academic documents through a web-based interface. After submission, the backend processes the input using specialized AI modules designed for different modalities. Video and image inputs are analyzed using convolutional neural networks (CNNs) to detect deepfake characteristics and visual inconsistencies. Academic documents are processed using natural language processing techniques and transformer-based models to estimate the probability that the text was

generated by artificial intelligence. In addition, student project documents are compared against existing submissions stored in the institutional database to calculate similarity scores and detect potential academic plagiarism.

The system also includes an automated CV generation module, which allows users to input personal and professional information and receive a structured, professional CV generated automatically by the system.

By integrating multiple AI capabilities into a centralized platform, the system aims to enhance academic integrity, support media authenticity verification, and provide practical digital tools for students and institutions.

Design Goals

The design goals of the system define the key principles that guide architectural decisions and system implementation.

1. Performance

Performance is an important consideration in the design of the Multi-modal AI Integrity Suite. The system is designed to efficiently process multiple types of digital content, including videos, images, and textual documents. Uploaded media files are analyzed through optimized processing pipelines that include data preprocessing, feature extraction, and AI model inference.

The backend system handles incoming requests using efficient API mechanisms, ensuring that file uploads and analysis tasks are processed in a timely manner. Parallel processing strategies are used in some modules, such as performing similarity analysis and AI text detection simultaneously, to improve response time and overall system efficiency. These optimizations allow the system to deliver analysis results within an acceptable time frame.

2. Dependability

Dependability ensures that the system operates reliably under different conditions and maintains consistent performance. The system is designed to handle various potential issues

such as invalid file uploads, unsupported formats, network interruptions, or AI model processing errors.

Proper exception handling and logging mechanisms are implemented throughout the backend system to monitor errors and ensure that failures are handled gracefully. User-friendly error messages guide users when issues occur, helping maintain system reliability and stability during normal usage.

3. Security

Security is a critical component of the system design because the platform processes sensitive data such as student documents and personal information. The system incorporates several security mechanisms to protect user data and maintain privacy.

User authentication mechanisms ensure that only authorized individuals can access system features. Passwords and sensitive user information are securely stored using encryption techniques. In addition, communication between the frontend interface and backend services is protected using secure communication protocols.

Uploaded media files and documents are handled securely within the processing environment to prevent unauthorized access or misuse. These security measures help build user trust and ensure that the system adheres to responsible data handling practices.

4. Scalability

The system is designed with scalability in mind to accommodate increasing user demand and future feature expansion. The architecture follows a modular design, allowing different components such as the frontend interface, backend services, and AI processing modules to scale independently.

This modular architecture makes it possible to add additional detection capabilities, such as new AI models or additional media analysis features, without requiring major changes to the existing system structure. As a result, the system can evolve over time to support new technologies and larger workloads.

5. Usability

Usability is a key design objective to ensure that users with different levels of technical expertise can interact with the system effectively. The system provides a web-based dashboard that presents a clear and intuitive workflow for uploading content, initiating analysis, and viewing results.

Detection results are displayed in an easily understandable format that includes classification labels and confidence scores. For example, users can see whether uploaded media is classified as real or manipulated, and the system provides probability values to support interpretation of the results.

The CV generation module is also designed with simplicity in mind, allowing users to easily enter personal information and generate professional documents without requiring advanced technical knowledge

6. Maintainability

Maintainability is achieved through a modular and well-organized system architecture that separates the frontend interface, backend services, AI processing modules, and database components. This separation of concerns allows developers to update or improve specific modules without affecting the entire system.

The use of standard development frameworks, structured coding practices, and version control systems further supports long-term maintenance and collaborative development. This design approach simplifies debugging, testing, and future enhancements, ensuring that the system can continue to evolve as new requirements and technologies emerge.

3.1 Architectural Design

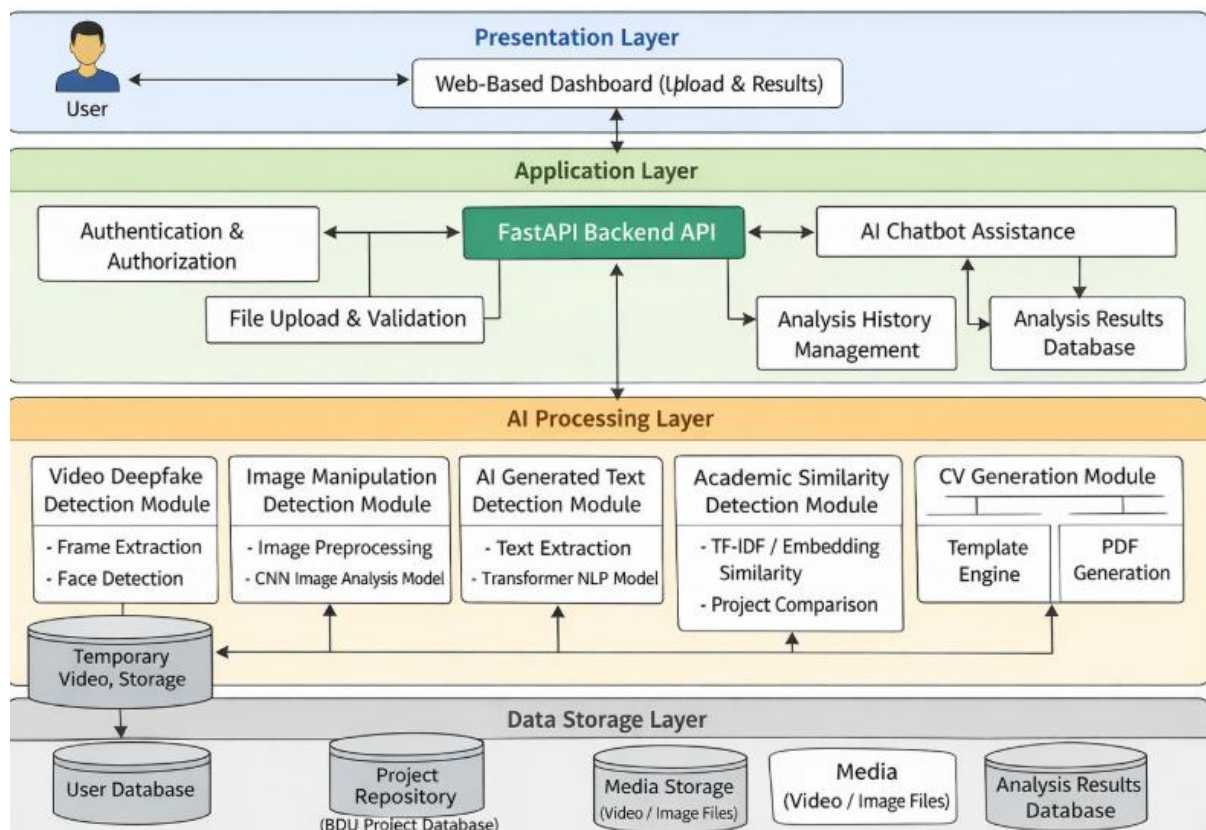


Figure 21: Architecture diagram

Architecture Design Description

The architecture of the Multi-modal AI Integrity Suite: Deepfake Detection, Content Authentication, and Intelligent Document Generation System is designed using a layered approach to ensure modularity, scalability, and efficient processing. The system is divided into four main layers: Presentation Layer, Application Layer, AI Processing Layer, and Data Storage Layer. Each layer performs a specific function and communicates with the other layers through well-defined interfaces. This layered architecture enables the system to process different types of digital content, including videos, images, and textual documents, while maintaining flexibility for future enhancements.

1. Presentation Layer

The Presentation Layer represents the user-facing component of the system. It consists of a web-based dashboard that allows users to interact with the system through a browser. Users

can register and log into the platform, upload videos, images, and academic documents for analysis, generate professional CVs, and view the results of authenticity verification.

This layer is responsible for collecting user inputs and displaying outputs generated by the backend services. The user interface provides clear workflows for uploading files, viewing analysis reports, and managing analysis history. Additionally, users can interact with the AI chatbot assistant, which provides guidance on system functionality and answers questions related to AI detection results.

Communication between the presentation layer and the backend services is performed through secure HTTP requests, ensuring reliable and secure interaction between users and the system.

2. Application Layer

The Application Layer serves as the central control layer of the system and is implemented using a FastAPI backend framework. This layer manages the core application logic and coordinates interactions between the user interface, AI processing modules, and data storage components.

The application layer includes several important services such as authentication and authorization, file upload and validation, analysis history management, and AI chatbot interaction. It ensures that uploaded media files and documents are validated before being processed by the AI modules. The backend also manages user sessions, handles API requests, and routes data to the appropriate processing modules depending on the type of content submitted.

For example, when a user uploads a video, image, or document, the FastAPI backend validates the input format and forwards the data to the corresponding AI analysis module within the AI processing layer. The application layer also manages the storage of analysis results and retrieves historical data for display within the user interface.

3. AI Processing Layer

The AI Processing Layer is responsible for performing all intelligent analysis tasks related to media authenticity verification and content evaluation. This layer contains several specialized AI modules designed to process different types of digital data.

For video inputs, the system performs video preprocessing, where frames are extracted and facial regions are identified using face detection techniques. These frames are then analyzed by a deepfake detection model that examines visual patterns and facial inconsistencies to determine whether the video has been manipulated.

For image inputs, an image manipulation detection module analyzes visual artifacts and irregularities using convolutional neural networks to determine whether the image has been altered or generated by artificial intelligence.

For academic documents, the system applies natural language processing techniques and transformer-based models to evaluate linguistic patterns and estimate the probability that the text was generated by AI. Additionally, a similarity detection module compares the submitted document with previously stored student projects using text similarity algorithms such as TF-IDF and embedding-based similarity analysis to identify potential academic overlap.

The system also includes a CV generation module, which processes structured user inputs and generates a professionally formatted curriculum vitae using predefined templates and automated document formatting.

The outputs produced by these AI modules include classification results, probability scores, similarity percentages, and generated documents, which are then forwarded to the application layer for storage and presentation.

4. Data Storage Layer

The Data Storage Layer manages all persistent data required by the system. This layer contains several storage components that support the operation of the platform.

The database stores user account information, uploaded file metadata, analysis results, similarity scores, and chatbot interaction logs. A project repository database stores previously submitted academic projects that are used for similarity comparison during academic integrity verification.

In addition, media storage systems are used to temporarily or permanently store uploaded videos and images before and after processing. Temporary storage may also be used during preprocessing stages such as frame extraction and intermediate analysis.

This layer ensures data integrity, secure storage, and efficient retrieval of information, enabling features such as viewing past analysis results and managing historical reports.

Overall System Workflow

The overall workflow of the system begins when a user accesses the web-based dashboard and uploads digital content such as a video, image, or academic document, or enters personal information for CV generation. The request is received by the FastAPI backend, which validates the input and forwards it to the appropriate module within the AI Processing Layer.

The selected AI module performs the required analysis, such as deepfake detection, image authenticity verification, AI text detection, academic similarity comparison, or CV generation. After processing is completed, the results are sent back to the application layer, where they are stored in the database and prepared for presentation.

Finally, the results are returned to the user through the web interface, where users can view authenticity classifications, similarity scores, probability estimates, or generated documents. Additional services such as chatbot interaction and analysis history management are integrated into this workflow to enhance usability and user support.

This layered architectural design improves system maintainability, scalability, and performance, while also enabling the integration of additional AI modules and future system enhancements without disrupting existing components.

3.1.1 Component Modeling

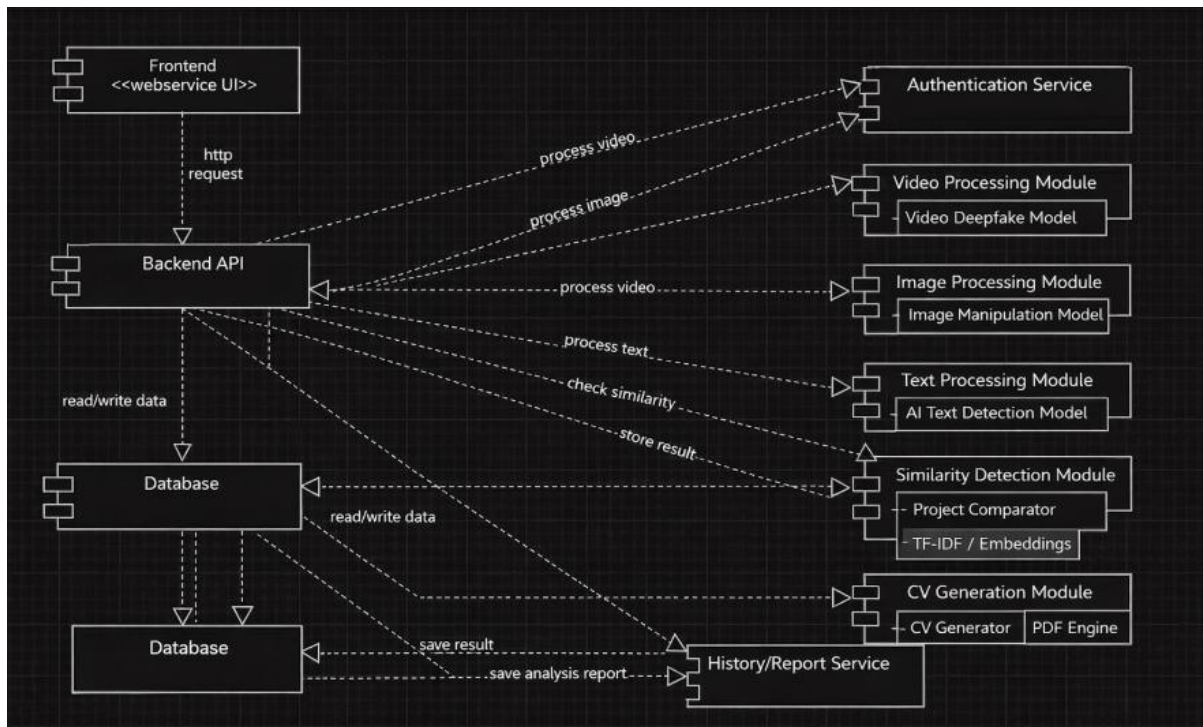


Figure 22: Component Modeling

Key Components

1. **Frontend (Web Service UI)**: The Frontend represents the user-facing web interface through which users interact with the system. It allows users to upload videos, images, and text, request analysis, check similarity, and generate CVs. All user actions are sent to the Backend API via HTTP requests, and results such as analysis outputs and reports are displayed back to the user.
2. **Backend API**: The Backend API acts as the central controller of the system. It receives requests from the frontend and routes them to the appropriate modules such as video processing, image processing, text analysis, similarity detection, and CV generation. It also manages communication with the database and ensures all services work together efficiently.
3. **Authentication Service**: This service handles user authentication and authorization. It verifies user credentials, manages login sessions, and ensures secure access to system features. It interacts with the database to store and retrieve user account information.

4. **Video Processing Module (Video Deepfake Model):** This module processes uploaded videos by extracting frames and preparing them for analysis. The integrated Deepfake Detection Model analyzes the processed video data to determine whether the content is real or manipulated, returning the results to the Backend API.
5. **Image Processing Module (Image Manipulation Model):** The Image Processing Module is responsible for analyzing uploaded images. It detects signs of manipulation using AI models and classifies images as authentic or fake. The processed results are sent back to the Backend API for further use and storage.
6. **Text Processing Module (AI Text Detection Model):** This module analyzes textual content to determine whether it is human-written or AI-generated. It uses NLP-based models to evaluate patterns in the text and returns classification results to the Backend API.
7. **Similarity Detection Module (Project Comparator – TF-IDF / Embeddings):** This module compares submitted text or project documents with existing data to identify similarities. It uses techniques such as TF-IDF and embeddings to compute similarity scores, helping detect plagiarism or duplicated academic work.
8. **CV Generation Module (CV Generator & PDF Engine):** The CV Generation Module allows users to input their personal and professional details and automatically generates a structured CV. The PDF Engine converts the generated CV into a downloadable document format.
9. **History/Report Service:** This service manages storage and retrieval of all analysis results. It keeps a record of user activities, including processed videos, images, text analysis, and similarity checks. It also generates reports that users can review or download later.
10. **Database:** The Database serves as the persistent storage layer of the system. It stores user data, analysis results, similarity records, and generated reports. All read and write operations are handled through the Backend API to ensure consistency and security.

Relationships and Interactions

- The Frontend communicates only with the Backend API through HTTP requests.
- The Backend API acts as a mediator and routes requests to:

- Authentication Service for user validation
- Video, Image, and Text Processing Modules for content analysis
- Similarity Detection Module for project comparison
- CV Generation Module for document creation
- The Processing Modules return results (analysis, classification, similarity scores) back to the Backend API.
- The Backend API stores results in the Database and forwards them to the History/Report Service for report generation.
- The History/Report Service manages saved results and enables users to retrieve past analyses.
- The Database supports all components by storing:
 - User credentials
 - Analysis outputs
 - Similarity results
 - Generated CVs and reports

This component structure follows a modular and scalable design, where each service is independent but coordinated through the Backend API. It clearly separates responsibilities such as authentication, processing, analysis, and reporting, making the system easier to maintain, extend, and deploy.

3.1.2 Deployment Modeling

The deployment model of the Multi-modal AI Integrity Suite defines how system components are distributed across physical and virtual nodes and how they interact to deliver system functionalities. The system follows a distributed, multi-tier web-based architecture that ensures scalability, modularity, and efficient processing of multi-modal data, including video, image, and text.

The architecture consists primarily of the Client Device, Web Server, and Database Server. The Client Device hosts the frontend interface, allowing users to interact with the system by uploading content, requesting analysis, and viewing results. The Web Server serves as the core processing unit, hosting the Backend API, authentication mechanisms, AI processing modules (video, image, and text), similarity detection services, CV generation module, and

report management services. The Database Server is responsible for persistent storage of user data, analysis results, similarity records, and generated reports.

Communication between these components is carried out using secure HTTP/HTTPS protocols, ensuring reliable and standardized data exchange. The Web Server interacts with the Database Server for all data storage and retrieval operations, while internal modules communicate seamlessly to process requests. This deployment structure supports centralized processing of AI models while maintaining a clear separation between user interaction, application logic, and data storage, thereby enhancing system performance, security, and maintainability.

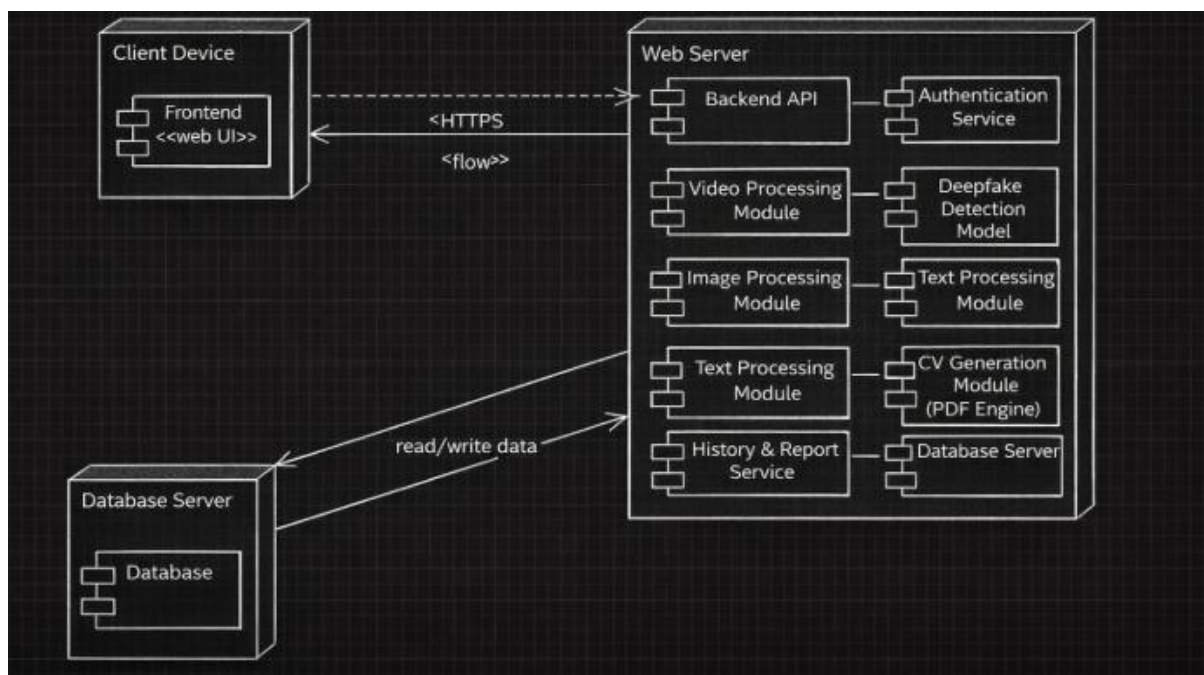


Figure : Deployment diagram

3.2 Detail Design

3.2.1 Design class model

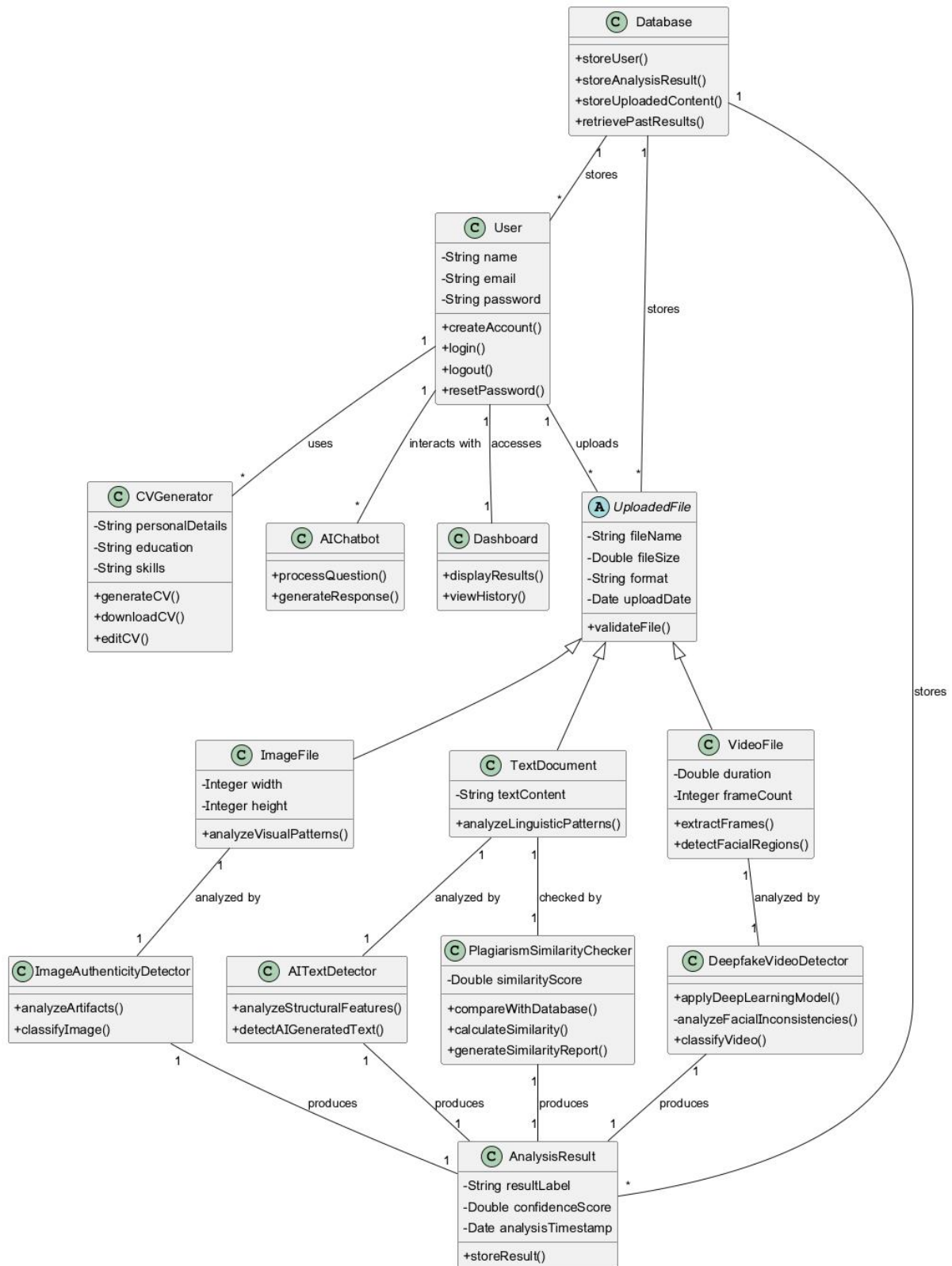


Figure 23: Design class model

3.2.2 Persistent Model

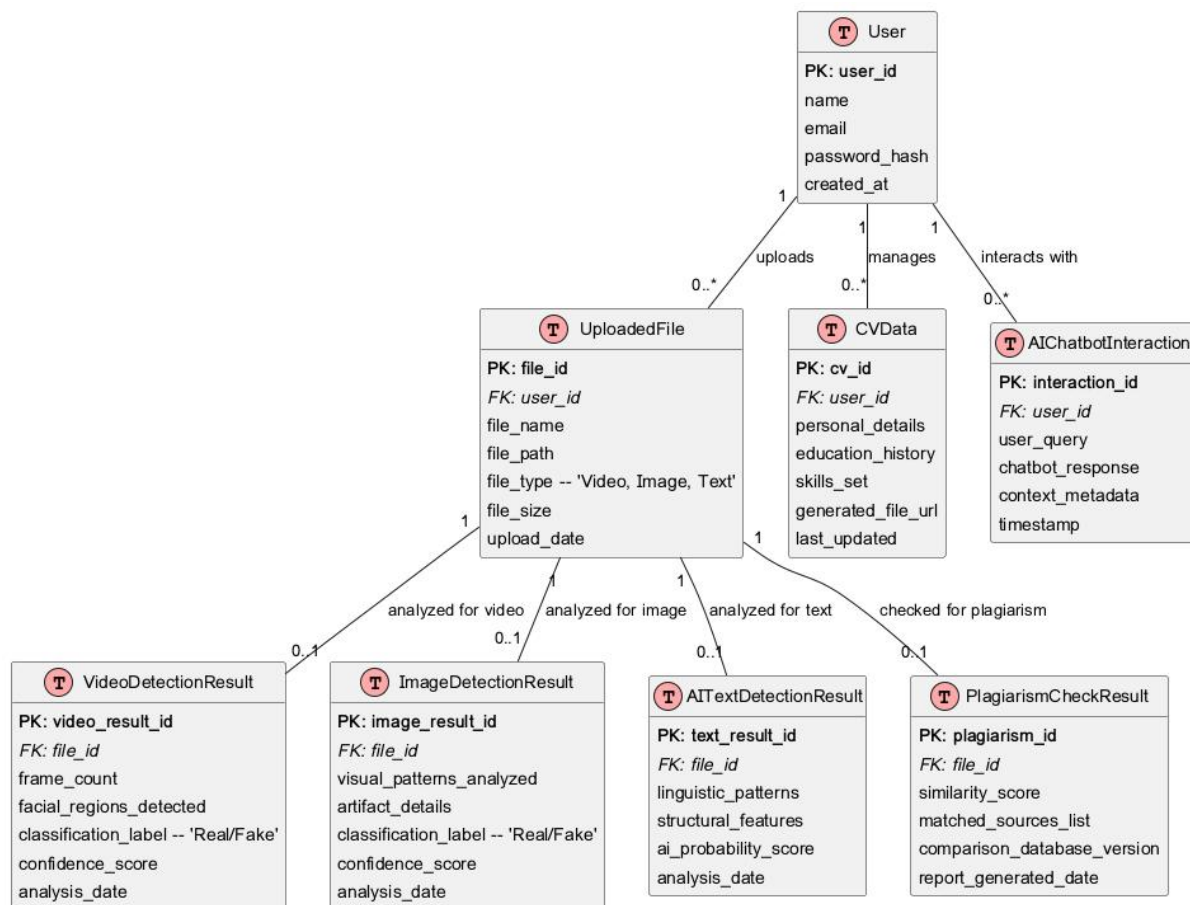


Figure 24: Persistent Model

This persistent model diagram represents how data is stored and managed in the Multi-modal AI Integrity Suite using an object-oriented approach. It shows the core entities of the system such as User, UploadedFile, VideoDetectionResult, ImageDetectionResult, AITextDetectionResult, PlagiarismCheckResult, CVDData, and AIChatbotInteraction, along with their attributes, data types, and key relationships.

A User can upload multiple UploadedFiles, which serve as the central repository for all digital content submitted for analysis. Each UploadedFile can then be associated with specific analysis results: a VideoDetectionResult for video content, an ImageDetectionResult for images, an AITextDetectionResult for textual content, and a PlagiarismCheckResult for academic documents. These result entities store detailed findings, confidence scores, and analysis dates, ensuring comprehensive traceability of all detection processes.

Furthermore, the model includes CVDData to store personal and professional information for Curriculum Vitae generation, directly linked to the User. All interactions with the AI-

powered chatbot are persistently logged in the AIChatbotInteraction entity, capturing user queries, chatbot responses, and timestamps to maintain conversational context and history. Overall, the diagram clearly illustrates data persistence, object relationships, multiplicity, and roles, ensuring efficient storage, traceability, and scalability of the multi-modal content analysis and user interaction processes.

3.3 User Interface Design

The user interface of the Multi-modal AI Integrity Suite is designed to be simple, intuitive, and user-friendly while supporting advanced AI-driven functionalities such as deepfake detection, text analysis, plagiarism checking, and intelligent document generation. The goal is to ensure that both standard users and researchers can efficiently interact with the system without requiring technical expertise.

The interface follows a consistent layout with clear navigation, interactive components, and real-time feedback to enhance usability and trust.

● Home Page

The home page serves as the main entry point of the system. It introduces the purpose of the platform and provides an overview of all available AI-powered tools, including:

- Image/Video Deepfake Detection
- Text Authenticity Detection
- CV Generator
- Plagiarism Checker
- AI Chatbot
- Community Challenge

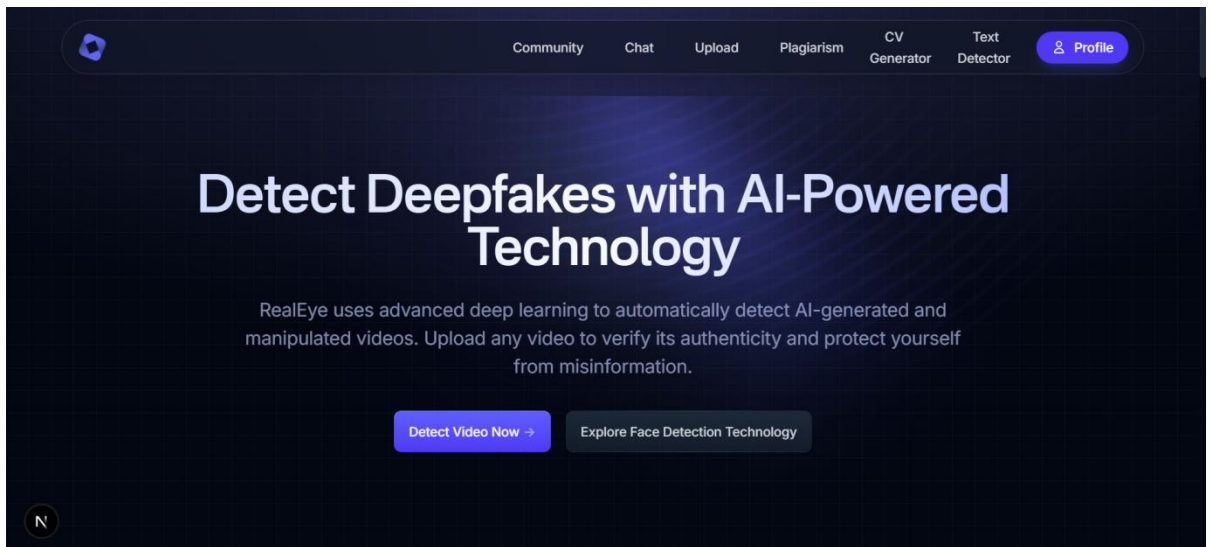


Figure 25: Home Page

• Login Page

The login page allows registered users to securely access the system by entering their credentials. The system validates user input and ensures secure authentication before

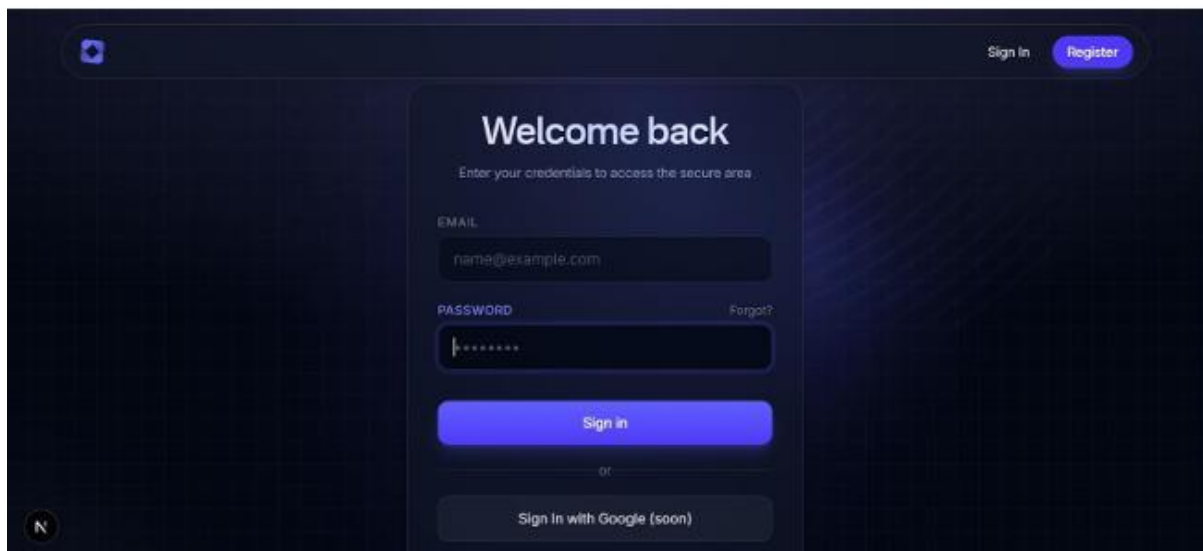


Figure 26: Login Page

• Register Page

The registration page enables new users to create an account by providing essential information such as username, email, and password.

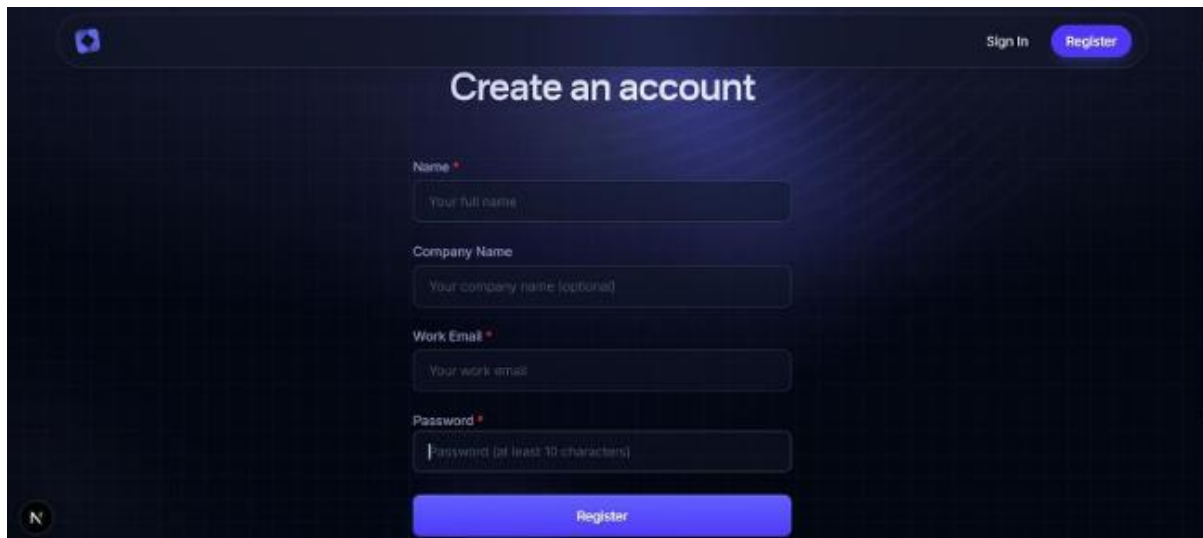


Figure 27: Register Page

● **Detection Page (Multi-modal Detection Interface)**

The detection page is the core functional interface of the system and has been expanded to support multi-modal AI detection capabilities.

Users can choose between different detection modes:

Image/Video Deepfake Detection

- Upload images or videos
- Preview uploaded content
- Detect whether media is real or fake
- Display confidence score (%)
- Highlight manipulated regions (if applicable)

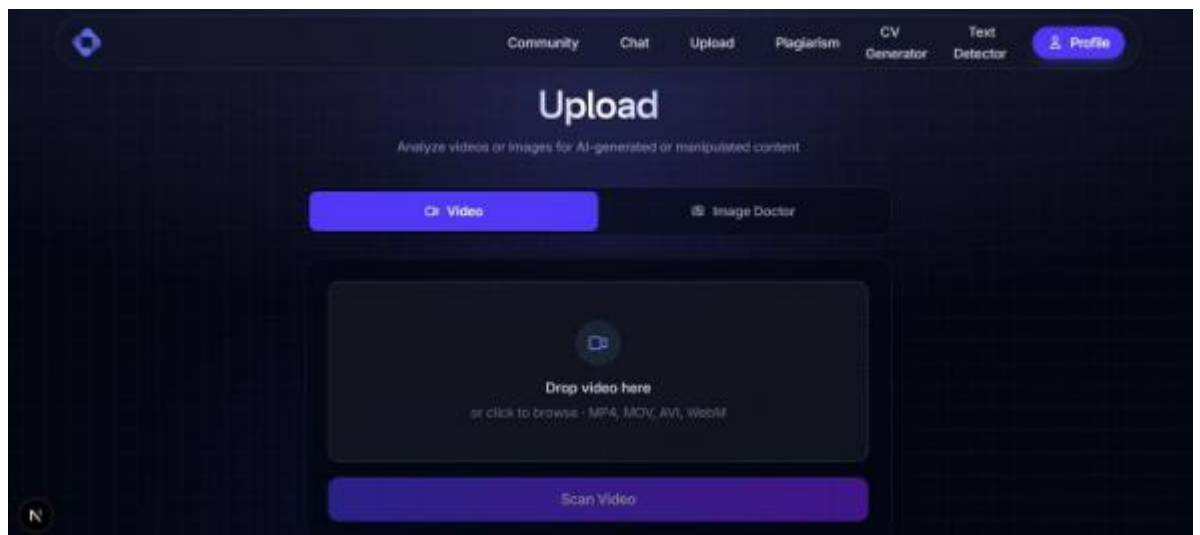


Figure 28: Detection Page (Multi-modal Detection Interface)

• CV Generator Page

The CV Generator page allows users to create professional, structured resumes using AI assistance.

Users can:

- Enter personal and professional details (education, skills, experience)
- Select from predefined CV templates
- Generate a formatted CV instantly
- Edit and download the document (PDF/DOC format)

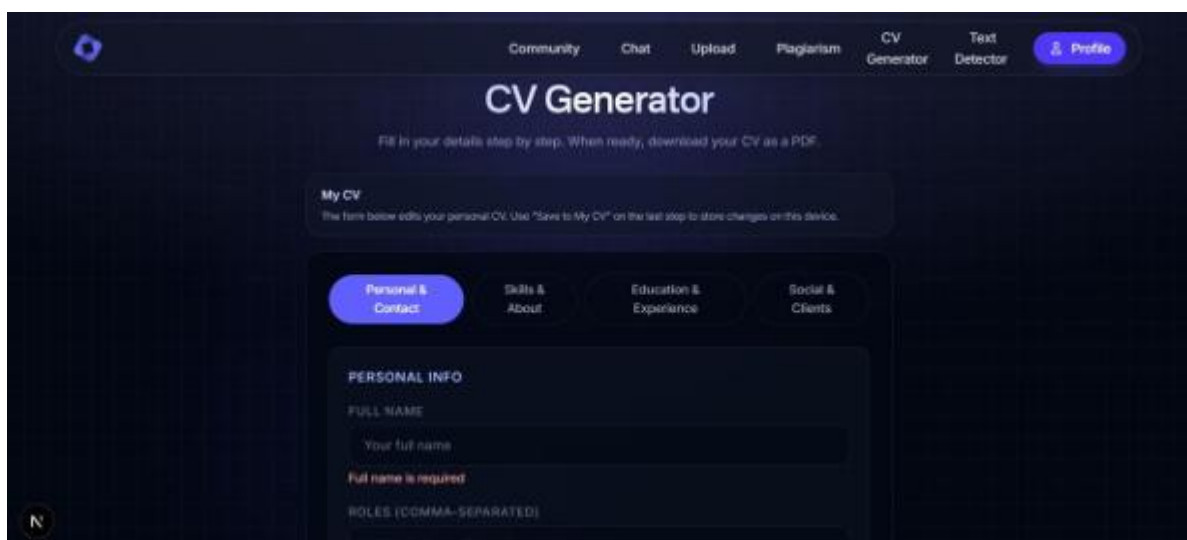


Figure 29: CV Generator Page

- **Plagiarism Checker Page**

The plagiarism checker enables users to verify the originality of textual content.

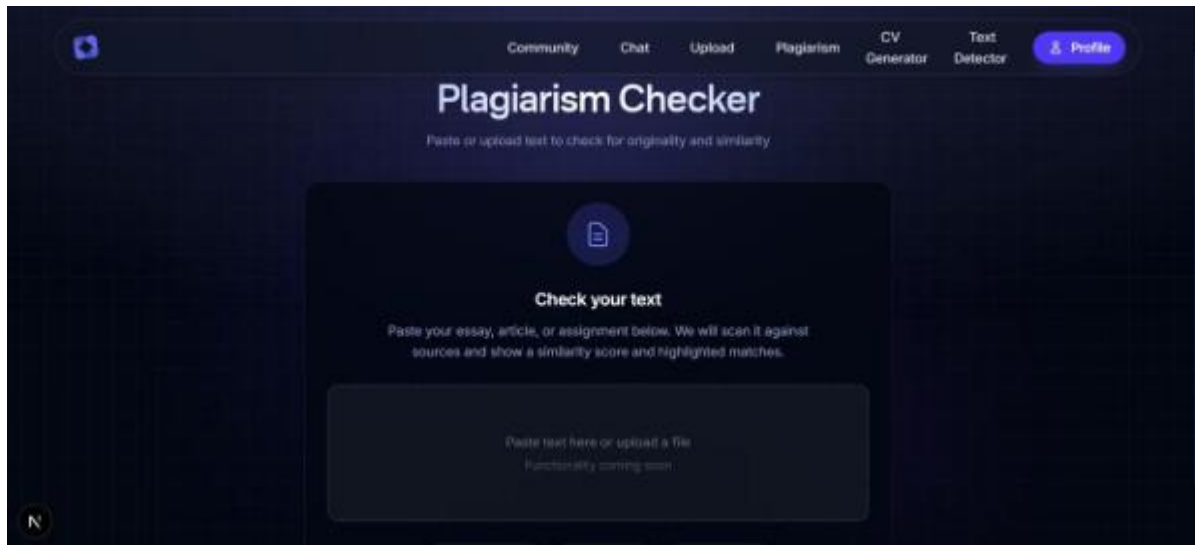


Figure 30: Plagiarism Checker Page

- **Chatbot Page**

The chatbot page provides intelligent and interactive assistance to users. It enhances user experience by offering real-time support and guidance.

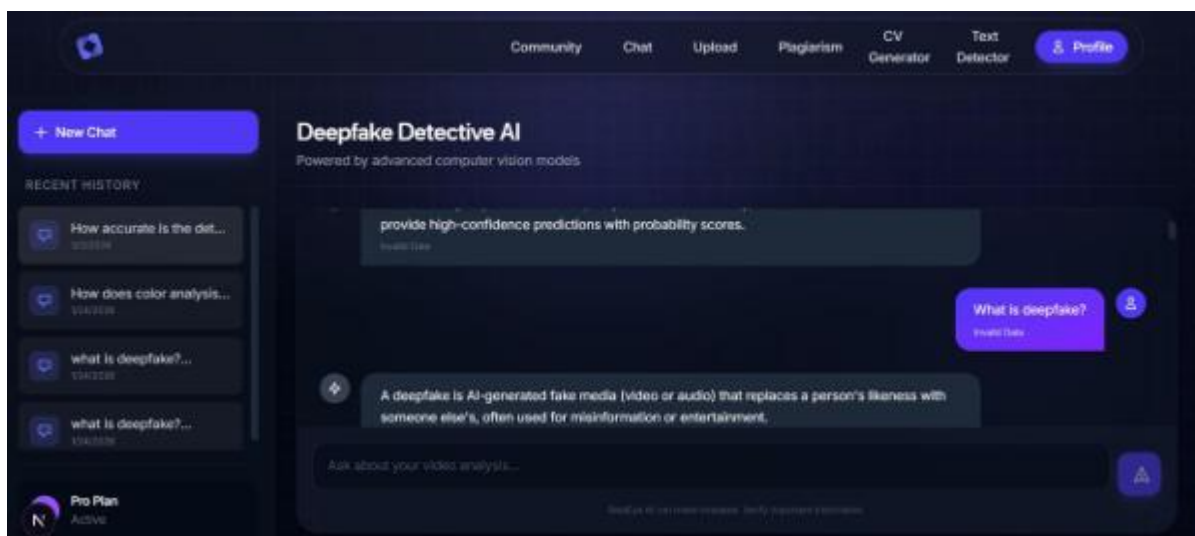


Figure 31: Chatbot Page

- **Community Challenge Page**

The community challenge page introduces an interactive environment where users can engage with real-world detection scenarios.



Figure 32: Community Challenge Page

3.4 Access Control and Security

This section describes the access control mechanisms and security measures implemented in the Multi-modal AI Integrity Suite, which integrates deepfake detection, text authenticity analysis, plagiarism checking, CV generation, and AI-assisted services. The system is designed to ensure that only authorized users can access specific functionalities while maintaining the confidentiality, integrity, and availability of sensitive data and AI resources.

3.4.1 Access Control

Access control is critical because the system handles sensitive resources, including multimedia files (images and videos), user-submitted text, generated documents (CVs), analysis results, and trained AI models. Without proper restrictions, these resources could be exposed, modified, or misused. Therefore, the system enforces strict authentication and role-based authorization mechanisms.

All users must authenticate using valid credentials before accessing the system. Passwords are securely stored using hashing techniques, and token-based authentication (such as JWT)

is used to manage user sessions. Once authenticated, each request made by the user is validated to ensure it is associated with a valid and active session.

User Roles

The system defines two primary roles to control access:

Researcher / Developer

This role is responsible for maintaining, monitoring, and improving the system's AI capabilities. Users in this role have access to advanced and technical functionalities, including:

- Uploading, organizing, and managing datasets (images, videos, and text samples)
- Running, retraining, and monitoring AI models for deepfake detection, text analysis, and plagiarism detection
- Evaluating model performance using metrics such as accuracy, precision, recall, and F1-score
- Accessing detailed analytical outputs, logs, and system reports
- Managing model versions, updates, and deployment processes

Although this role has broader access, it is still restricted from critical system-level configurations such as authentication mechanisms or core security settings to prevent accidental or malicious system compromise.

Standard User (End User)

This role is designed for general system usage with limited and controlled access. Standard users can interact with all user-facing features without accessing internal system components. Their permissions include:

- Uploading and analyzing images/videos for deepfake detection
- Entering or pasting text for AI-authenticity detection
- Using the plagiarism checker to verify content originality
- Generating, editing, and downloading CVs
- Interacting with the AI chatbot for guidance and assistance
- Participating in community challenges

- Viewing and managing their personal analysis history and results

Standard users are restricted from accessing:

- Training datasets and raw data collections
- AI model parameters, architectures, and training processes
- Intermediate processing outputs such as feature vectors
- System configuration and administrative settings

This separation ensures that sensitive system components remain protected while maintaining a smooth and user-friendly experience.

Authorization and Enforcement

After successful authentication, the system applies authorization by verifying the user's role before executing any requested action. Each request is checked against predefined permissions, and any unauthorized attempt is immediately rejected.

Access control is enforced at multiple layers:

- Application Level: All API endpoints validate the user's authentication token and role before processing requests
- Database Level: Access to sensitive data such as datasets, models, and logs is restricted based on role permissions

This layered approach ensures that even if one level is compromised, other layers continue to protect the system.

Data Protection and Monitoring

To maintain data security and system reliability, several protection mechanisms are implemented:

- User credentials are securely stored using hashing algorithms
- All communication between client and server is encrypted using HTTPS
- Uploaded files (images, videos, and documents) are stored in controlled environments with restricted access

- Generated outputs such as CVs and analysis reports are accessible only to the respective user
- System-level data (models, datasets) is separated from user-level data to prevent unauthorized exposure

The system also includes logging and monitoring to track critical activities, including:

- User login and logout attempts
- File uploads and analysis requests (deepfake detection, text analysis, plagiarism checks)
- Document generation activities (CV creation and downloads)
- Model updates and system-level operations

3.4.2 Security Measures

The Multi-modal AI Integrity Suite implements multiple layers of security to ensure the confidentiality, integrity, and availability of its data and services. To secure user access, the system employs strong authentication mechanisms in which user passwords are never stored in plain text but are instead hashed using robust cryptographic algorithms such as bcrypt, often combined with salting for additional protection. In addition, token-based authentication, such as JSON Web Tokens (JWT), is used to manage user sessions securely. Once a user is authenticated, a token is issued and must be included in all subsequent requests, ensuring that only verified users can access protected resources.

All communication between the client and server is secured using HTTPS, which encrypts transmitted data and protects sensitive information such as login credentials, uploaded media files, and analysis results from interception or tampering. Furthermore, the system enforces strict role-based access control, where each user is assigned a specific role and permissions are applied accordingly. Every request is validated against the user's role to ensure that access to functionalities and data remains restricted to authorized users only. Sensitive data, including trained AI models, datasets, user-uploaded content, and generated documents such as CVs, is stored in protected environments with controlled access. Proper storage-level permissions are enforced to prevent unauthorized retrieval or modification. In addition, the system applies strict file validation procedures for all uploads, ensuring that files meet

defined type and size requirements before processing. This reduces the risk of malicious file injection and protects system resources from abuse.

To further enhance security, all user inputs are validated and sanitized before being processed. This prevents common vulnerabilities such as injection attacks and ensures that only safe and expected data is handled by the system. The system also implements rate limiting to control the number of requests that can be made within a specific time frame, thereby reducing the risk of denial-of-service attacks and preventing excessive resource consumption. Finally, regular system maintenance and updates are considered an integral part of the security strategy. All dependencies, libraries, and frameworks are kept up to date to minimize vulnerabilities and ensure that the system remains resilient against emerging security threats.

3.4.3 Security Workflow Example

The security workflow of the Multi-modal AI Integrity Suite demonstrates how authentication, authorization, and data protection mechanisms are continuously applied during user interaction. The process begins when a user attempts to log into the system by providing valid credentials. The system verifies these credentials by comparing the hashed version of the entered password with the stored hash in the database. If the credentials are valid, a secure authentication token is generated and issued to the user; otherwise, access is immediately denied. Once authentication is successful, the system identifies the user's role and establishes a secure session. The issued authentication token is then attached to all subsequent requests, ensuring that each action performed by the user is associated with a verified identity. Whenever a user initiates a request, such as uploading a video for deepfake detection, submitting text for analysis, generating a CV, or performing a plagiarism check, the system first validates the token to ensure that it is valid and has not expired. Requests with invalid or missing tokens are rejected, preventing unauthorized access.

After validating the request, the system performs an authorization check by comparing the requested action with the user's assigned role. Standard users are permitted to access general functionalities such as detection tools, document generation, and chatbot services, while restricted actions involving datasets, models, or internal configurations are denied. Researchers and developers, on the other hand, are granted access to advanced operations within defined limits. This step ensures strict enforcement of role-based access control and prevents privilege escalation. If the request is authorized, the system proceeds with secure

processing. Uploaded files are first validated to ensure they meet required formats and size constraints. The relevant AI models then process the request, whether for deepfake detection, text analysis, or plagiarism checking. During this stage, sensitive intermediate data, such as extracted features or internal computations, is protected and not exposed to unauthorized users.

Once processing is complete, the system generates the appropriate response, such as detection results, plagiarism scores, or generated CV documents. Only the information permitted for the user's role is included in the response, which is then transmitted securely through encrypted communication channels. Throughout the entire process, the system maintains detailed logs of critical activities, including login attempts, file uploads, analysis requests, and system operations. These logs support monitoring, auditing, and the detection of suspicious behavior, ensuring that the system remains secure, reliable, and accountable.

Reference

- [1] <https://rsisinternational.org/journals/ijrias/articles/deepfake-detection-using-multimodal-ai/>
- [2] https://www.researchgate.net/publication/395360891_Deepfake_Detection_A_Multimodal_Survey
- [3] <https://www.frontiersin.org/journals/artificialintelligence/articles/10.3389/frai.2025.1665798/full>
- [4] <https://www.frontiersin.org/journals/artificialintelligence/articles/10.3389/frai.2026.1737790/full>